Report-cell

S.No	Reference	Library	Unit Area	Cell Count	Total Area	Attributes
1	AND2X1_LVT	saed32rvt_ttftp7ltvs40c	2.033152	253	514.387482	
2	AND2X1_RVT	saed32rvt_ttftp7ltvs40c	2.033152	2	4.066304	
3	AND2X2_LVT	saed32rvt_ttftp7ltvs40c	2.287296	1	2.287296	
4	AND3X1_LVT	saed32rvt_ttftp7ltvs40c	2.287296	38	86.917250	
5	AND3X2_LVT	saed32rvt_ttftp7ltvs40c	2.541440	1	2.541440	
6	AND4X1_LVT	saed32rvt_ttftp7ltvs40c	2.541440	19	48.287360	
7	AND4X1_RVT	saed32rvt_ttftp7ltvs40c	2.541440	1	2.541440	
8	AND4X2_LVT	saed32rvt_ttftp7ltvs40c	3.812160	1	3.812160	
9	AO21X1_LVT	saed32rvt_ttftp7ltvs40c	2.541440	103	261.768321	
10	AO22X1_LVT	saed32rvt_ttftp7ltvs40c	2.541440	1116	2836.247051	
11	AO22X1_RVT	saed32rvt_ttftp7ltvs40c	2.541440	3	7.624320	
12	AO221X1_LVT	saed32rvt_ttftp7ltvs40c	3.049728	171	521.503474	
13	AO221X1_RVT	saed32rvt_ttftp7ltvs40c	3.049728	30	91.491838	
14	AO221X2_LVT	saed32rvt_ttftp7ltvs40c	3.303872	32	105.723907	
15	AO222X1_LVT	saed32rvt_ttftp7ltvs40c	3.303872	214	707.028631	
16	AO222X1_RVT	saed32rvt_ttftp7ltvs40c	3.303872	1	3.303872	
17	AO31X1_LVT	saed32rvt_ttftp7ltvs40c	3.049728	5	15.248640	
18	AO22X1_LVT	saed32rvt_ttftp7ltvs40c	3.558016	24	85.392385	
19	AOI22X1_LVT	saed32rvt_ttftp7ltvs40c	3.812160	24	91.491840	
20	DFFX1_LVT	saed32rvt_ttftp7ltvs40c	6.607744	1229	8120.917643	n
21	DFFX1_RVT	saed32rvt_ttftp7ltvs40c	6.607744	205	1354.587564	n
22	DFFX2_LVT	saed32rvt_ttftp7ltvs40c	7.116032	32	227.713024	n
23	INVX0_LVT	saed32rvt_ttftp7ltvs40c	1.270720	7	8.895040	
24	INVX0_RVT	saed32rvt_ttftp7ltvs40c	1.270720	3	3.812160	
25	INVX1_LVT	saed32rvt_ttftp7ltvs40c	1.270720	595	756.078403	
26	INVX2_LVT	saed32rvt_ttftp7ltvs40c	1.524864	9	13.723776	
27	INVX4_LVT	saed32rvt_ttftp7ltvs40c	2.033152	8	16.265217	
28	LATCHX1_LVT	saed32rvt_ttftp7ltvs40c	5.082880	63	320.221441	n
29	LATCHX1_RVT	saed32rvt_ttftp7ltvs40c	5.082880	5	25.414400	n
30	MUX21X1_LVT	saed32rvt_ttftp7ltvs40c	3.303872	59	194.928454	
31	MUX21X1_RVT	saed32rvt_ttftp7ltvs40c	3.303872	230	759.890985	
32	NAND2X0_LVT	saed32rvt_ttftp7ltvs40c	1.524864	135	205.856634	
33	NAND2X0_RVT	saed32rvt_ttftp7ltvs40c	1.524864	2	3.049728	
34	NAND2X2_LVT	saed32rvt_ttftp7ltvs40c	2.795584	1	2.795584	
35	NAND3X0_LVT	saed32rvt_ttftp7ltvs40c	1.779008	66	117.414530	
36	NAND3X0_RVT	saed32rvt_ttftp7ltvs40c	1.779008	3	5.337024	
37	NAND4X0_LVT	saed32rvt_ttftp7ltvs40c	2.033152	115	233.812492	
38	NR0FFX2_LVT	saed32rvt_ttftp7ltvs40c	2.033152	15	30.497282	
39	NR0FFX2_RVT	saed32rvt_ttftp7ltvs40c	2.033152	2	4.066304	
40	NOR2X0_LVT	saed32rvt_ttftp7ltvs40c	2.541440	7	17.790080	
41	NOR3X0_LVT	saed32rvt_ttftp7ltvs40c	2.795584	5	13.977920	
42	NOR4X1_LVT	saed32rvt_ttftp7ltvs40c	3.049728	12	36.596735	
43	OA21X1_LVT	saed32rvt_ttftp7ltvs40c	2.541440	13	33.038720	
44	OA22X1_LVT	saed32rvt_ttftp7ltvs40c	2.541440	43	109.281920	
45	OA221X1_LVT	saed32rvt_ttftp7ltvs40c	3.049728	167	509.304062	
46	OA222X1_LVT	saed32rvt_ttftp7ltvs40c	3.303872	160	528.619537	
47	OAI21X1_LVT	saed32rvt_ttftp7ltvs40c	3.049728	9	27.447551	
48	OAI21X1_RVT	saed32rvt_ttftp7ltvs40c	3.049728	31	94.541565	
49	OAI22X1_LVT	saed32rvt_ttftp7ltvs40c	3.049728	17	51.845375	
50	OAI22X1_RVT	saed32rvt_ttftp7ltvs40c	3.558016	3	10.674048	
51	OAI222X1_RVT	saed32rvt_ttftp7ltvs40c	3.812160	1	3.812160	
52	OR2X1_LVT	saed32rvt_ttftp7ltvs40c	2.033152	16	32.530434	
53	OR3X1_LVT	saed32rvt_ttftp7ltvs40c	2.287296	4	9.149184	
54	OR4X1_LVT	saed32rvt_ttftp7ltvs40c	3.558016	15	53.370241	
55	XNOR2X1_LVT	saed32rvt_ttftp7ltvs40c	4.320448	1	4.320448	
56	XNOR2X1_RVT	saed32rvt_ttftp7ltvs40c	4.320448	1	4.320448	
57	XOR2X1_RVT	saed32rvt_ttftp7ltvs40c	4.320448	1	4.320448	
58	picorv32_pcpi_fast_mul			7840	596464	h, n
59	picorv32a_DW01_add_0			156	044411	h
60	picorv32a_DW01_add_1			151	215675	h
61	picorv32a_DW01_add_2			182	983680	h
62	picorv32a_DW01_add_3			182	895296	h
63	picorv32a_DW01_sub_4			156	044411	h
64	picorv32a_DW01_add_5			156	044411	h
65	picorv32a_DW01_add_6			188	066561	h
66	picorv32a_DW01_cmpl_0			199	248097	h
67	picorv32a_DW01_dec_0_DW01_d			181	458819	h
68	picorv32a_DW01_inc_0			210	431229	h
69	picorv32a_DW01_inc_1			210	431229	h
70	picorv32a_DW01_sub_0			196	707451	h
71	picorv32a_DW_cmp_0			242	961665	h
	Total top module cells				**5343**	
	Total no of cells				**8598**	

Report_Area

S.No	Number of	Count	Area
1	**Number of ports:**	1819	
2	**Number of nets:**	11618	
3	**Number of cells:**	8546	
4	**Number of combinational cells:**	6806	
5	**Number of sequential cells:**	1667	
6	**Number of macros/black boxes:**	0	
7	**Number of buf/inv:**	1143	
8	**Number of references:**	71	
9	**Combinational area:**		18679.329838
10	**Buf/inv area:**		1580.521542
11	**Noncombinational area:**		10927.684057
12	**Macro/Black Box area:**		0.000000
13	**Net Interconnect area:**		8977.222396
	Total Cell Area		**29607.013886**
	Total Area		**38584.236291**

Report_power-1

S.No	Design	Wire Load Model	Library
1	**picorv32a**	35000	saed32rvt_ttftp7ltvs4
2	**picorv32_pcpi_fast_mul**	8000	saed32rvt_ttftp7ltvs4
3	**picorv32a_DW01_add_0**	ForQA	saed32rvt_ttftp7ltvs4
4	**picorv32a_DW01_add_1**	ForQA	saed32rvt_ttftp7ltvs4
5	**picorv32a_DW01_inc_0**	8000	saed32rvt_ttftp7ltvs4
6	**picorv32a_DW01_add_2**	ForQA	saed32rvt_ttftp7ltvs4
7	**picorv32a_DW01_inc_1**	8000	saed32rvt_ttftp7ltvs4
8	**picorv32a_DW01_add_3**	ForQA	saed32rvt_ttftp7ltvs4
9	**picorv32a_DW_cmp_0**	8000	saed32rvt_ttftp7ltvs4
10	**picorv32a_DW01_add_4**	ForQA	saed32rvt_ttftp7ltvs4
11	**picorv32a_DW01_sub_0**	ForQA	saed32rvt_ttftp7ltvs4
12	**picorv32a_DW01_add_5**	ForQA	saed32rvt_ttftp7ltvs4
13	**picorv32a_DW01_dec_0_DW01_dec_1**	ForQA	saed32rvt_ttftp7ltvs4
14	**picorv32a_DW01_cmpl_0**	ForQA	saed32rvt_ttftp7ltvs4
15	**picorv32_pcpi_fast_mul_DW_mult_tc_0**	8000	saed32rvt_ttftp7ltvs4
16	**picorv32a_DW01_add_6**	ForQA	saed32rvt_ttftp7ltvs4

Cell Internal Power = 1.7865 mW (100%)
Net Switching Power = 6.6951 uW (0%)

Total Dynamic Power = 1.7932 mW (100%)
Cell Leakage Power = 3.1202 mW

Report-power-2

S.No	Power Group	Internal Power	Switching Power	Leakage Power	Total Power	Percentage	Attributes
1	**io_pad**	0.0000	0.0000	0.0000	0.0000	0.00%	
2	**memory**	0.0000	0.0000	0.0000	0.0000	0.00%	
3	**black_box**	0.0000	0.0000	0.0000	0.0000	0.00%	
4	**clock_network**	1.7812E-03	0.0000	0.0000	1.7812E-03	36.25%	i
5	**register**	-7.2144E-01	1.1226	1.9063E+09	1.9063E+03	38.81%	
6	**sequential**	-2.7243E-01	0.1032	7.4802E+07	74.6327	1.52%	
7	**combinational**	6.3211	5.4694	1.1391E+09	1.1508E+03	23.42%	
	Total (uW)	**1.7865E+03**	**6.6952**	**3.1202e+09 pW**	**4.9134E+03**		
	Total Dynamic Power	**3.1202 mW**					

Report-wireload model (Compile)

Wire load model	35000
Location	**picorv32a (design)**
Resistance	**0.001107**
Capacitance	**0.00027**
Area	**0.01**
Slope	**140.36**

Report-Wire load Model 35000 (Compile)

S.No	Fanout	Length	Points	Average cap	Standard Deviation	%SD
1	1			16.92		
2	2			39.03		
3	3			64.09		
4	4			92.51		
5	5			124.73		
6	6			161.17		
7	7			202.25		
8	8			248.42		
9	9			300.09		
10	10			357.68		
11	11			421.63		
12	12			492.27		
13	13			570.31		
14	14			655.89		
15	15			749.54		
16	16			851.67		
17	17			962.72		
18	18			1083.11		
19	19			1213.28		
20	20			1353.64		
Total Length						

Report-Wire load Model -8000 (Compile)

S.No	Fanout	Length	Points	Average cap	Standard Deviation	%SD
1	1			13.94		
2	2			31.80		
3	3			51.61		
4	4			73.61		
5	5			98.05		
6	6			125.17		
7	7			155.23		
8	8			188.42		
9	9			225.12		
10	10			265.45		
11	11			309.71		
12	12			358.13		
13	13			410.96		
14	14			468.46		
15	15			530.86		
16	16			598.42		
17	17			671.38		
18	18			749.98		
19	19			834.49		
20	20			925.13		
Total Length						

Report-Wire load Model -2 (Compile)

Wire load model:	8000
Location	**picorv32_pcpi_fast_mul (design)**
Resistance :	**0.0015727**
Capacitance :	**0.000312**
Area :	**0.01**
Slope :	**90.6464**

Report-Wire load Model -FORQA(Compile)

S.No	Fanout	Length	Points	Average cap	Standard Deviation	%SD
1	1			8.28		
2	2			18.49		
3	3			29.35		
4	4			40.92		
5	5			53.23		
6	6			66.36		
7	7			80.36		
8	8			95.27		
9	9			111.17		
10	10			128.09		
11	11			146.10		
12	12			165.26		
13	13			185.61		
14	14			207.22		
15	15			230.13		
16	16			254.41		
17	17			280.11		
18	18			307.28		
19	19			335.98		
20	20			366.27		
Total Length						

Report-Wire load Model -3 (Compile)

Wire load model:	ForQA
Location :	**picorv32a_DW01_add_0 (design)**
Resistance :	**0.002067**
Capacitance :	**0.026724**
Area :	**0.01**
Slope :	**30.3864**

Report-qor (Compile)

S.No	Timing Path Group (none)	
1	Levels of Logic:	70.00
2	Critical Path Length:	6.36
3	Critical Path Slack:	-1.43
4	Critical Path Clk Period:	6.00
5	Total Negative Slack:	-13.62
6	No. of Violating	19.00
7	Worst Hold	0.00
8	Total Hold Violation:	0.00
9	No. of Hold	0.00
	Cell Count	
10	Hierarchical Cell	15
11	Hierarchical Port	1510
12	Leaf Cell Count:	8473
13	BufInv Cell Count:	1143
14	Buf Cell Count:	26
15	Inv Cell Count:	1117
16	CT Buf/Inv Cell	0
17	Combinational Cell Count:	6806
18	Sequential Cell Count:	1667
19	Macro Count:	0
	Area	
20	Combinational Area:	18679.329838
21	Noncombinational Area:	10927.684057
22	BufInv Area:	1580.521542
23	Total Buffer Area:	98.61
24	Total Inverter Area:	1481.91
25	Macro/Black Box	0.000000
26	Net Area:	8977.222396
	Cell Area:	**29607.013886**
	Design Area:	**38584.236291**
	Design Rules	
	Total Number of Nets:	10245
	Nets With Violations:	0
	Max Trans	0
	Max Cap Violations:	0

Report_timing - 1 (Compile)

S.No	
1	**Startpoint:** gentbk1.pcpi_mul/n2_reg[1] (rising edge-triggered flip-flop clocked
2	**Endpoint:** gentbk1.pcpi_mul/n0_reg[63] (rising edge-triggered flip-flop clocked by clk)
3	**Path Group:** clk
4	**Path Type:** Max

Report_timing-2 (Compile)

Des/Clust/Port	Wire Load Model	Library
picorv32a	**35000**	**saed32rvt_ttftp7ltvs40c**
picorv32_pcpi_fast_mul	**8000**	**saed32rvt_ttftp7ltvs40c**
picorv32_pcpi_fast_mul_DW_mult_tc_0	**8000**	**saed32rvt_ttftp7ltvs40c**

report_timing(Compile)

S.No	Point	Incr	Path A
1	**clock clk (rise edge)**	0.00	0.00
2	**clock network delay (ideal)**	1.00	1.00
3	**gentbk1.pcpi_mul/n2_reg[1]/CLK (DFFX1_LVT)**	0.00 #	1.00 r
4	**gentbk1.pcpi_mul/n2_reg[1]/Q (DFFX1_LVT)**	0.11	1.11 r
5	**gentbk1.pcpi_mul/n2_2209/n[1] (picorv32_pcpi_fast_mul_DW_mult_tc_0)**	0.00	1.11 r
6	**gentbk1.pcpi_mul/mul_2209/U255/Y1 (HADDX2_LVT)**	0.07	1.18 r
7	**gentbk1.pcpi_mul/mul_2209/U251/Y (NAND2X0_LVT)**	0.03	1.22 f
8	**gentbk1.pcpi_mul/mul_2209/U254/Y (NAND2X0_LVT)**	0.05	1.27 r
9	**gentbk1.pcpi_mul/mul_2209/U273/CO (FADDX1_LVT)**	0.10	1.36 r
10	**gentbk1.pcpi_mul/mul_2209/U272/CO (FADDX1_LVT)**	0.09	1.46 r
11	**gentbk1.pcpi_mul/mul_2209/U271/CO (FADDX1_LVT)**	0.09	1.55 r
12	**gentbk1.pcpi_mul/mul_2209/U270/CO (FADDX1_LVT)**	0.09	1.65 r
13	**gentbk1.pcpi_mul/mul_2209/U269/CO (FADDX1_LVT)**	0.09	1.74 r
14	**gentbk1.pcpi_mul/mul_2209/U268/CO (FADDX1_LVT)**	0.09	1.84 r
15	**gentbk1.pcpi_mul/mul_2209/U267/CO (FADDX1_LVT)**	0.09	1.93 r
16	**gentbk1.pcpi_mul/mul_2209/U266/CO (FADDX1_LVT)**	0.09	2.03 r
17	**gentbk1.pcpi_mul/mul_2209/U265/CO (FADDX1_LVT)**	0.09	2.12 r
18	**gentbk1.pcpi_mul/mul_2209/U264/CO (FADDX1_LVT)**	0.09	2.22 r
19	**gentbk1.pcpi_mul/mul_2209/U263/S (FADDX1_LVT)**	0.15	2.37 f
20	**gentbk1.pcpi_mul/mul_2209/U276/Y (AOI21X1_LVT)**	0.10	2.47 f
21	**gentbk1.pcpi_mul/mul_2209/U275/Y (XOR2X1_LVT)**	0.10	2.57 r
22	**gentbk1.pcpi_mul/mul_2209/U313/CO (FADDX1_LVT)**	0.11	2.68 r
23	**gentbk1.pcpi_mul/mul_2209/U312/CO (FADDX1_LVT)**	0.09	2.77 r
24	**gentbk1.pcpi_mul/mul_2209/U313/CO (FADDX1_LVT)**	0.09	2.87 r
25	**gentbk1.pcpi_mul/mul_2209/U312/CO (FADDX1_LVT)**	0.09	2.96 r
26	**gentbk1.pcpi_mul/mul_2209/U311/CO (FADDX1_LVT)**	0.09	3.06 r
27	**gentbk1.pcpi_mul/mul_2209/U310/CO (FADDX1_LVT)**	0.09	3.15 r
28	**gentbk1.pcpi_mul/mul_2209/U309/CO (FADDX1_LVT)**	0.09	3.25 r
29	**gentbk1.pcpi_mul/mul_2209/U308/CO (FADDX1_LVT)**	0.09	3.34 r
30	**gentbk1.pcpi_mul/mul_2209/U307/CO (FADDX1_LVT)**	0.09	3.44 r
31	**gentbk1.pcpi_mul/mul_2209/U306/CO (FADDX1_LVT)**	0.09	3.53 r
32	**gentbk1.pcpi_mul/mul_2209/U305/CO (FADDX1_LVT)**	0.09	3.63 r
33	**gentbk1.pcpi_mul/mul_2209/U304/CO (FADDX1_LVT)**	0.09	3.72 r
34	**gentbk1.pcpi_mul/mul_2209/U303/CO (FADDX1_LVT)**	0.09	3.82 r
35	**gentbk1.pcpi_mul/mul_2209/U302/CO (FADDX1_LVT)**	0.09	3.91 r
36	**gentbk1.pcpi_mul/mul_2209/U301/CO (FADDX1_LVT)**	0.09	4.01 r
37	**gentbk1.pcpi_mul/mul_2209/U300/CO (FADDX1_LVT)**	0.09	4.10 r
38	**gentbk1.pcpi_mul/mul_2209/U299/CO (FADDX1_LVT)**	0.09	4.20 r
39	**gentbk1.pcpi_mul/mul_2209/U298/CO (FADDX1_LVT)**	0.09	4.29 r
40	**gentbk1.pcpi_mul/mul_2209/U297/CO (FADDX1_LVT)**	0.09	4.38 r
41	**gentbk1.pcpi_mul/mul_2209/U296/CO (FADDX1_LVT)**	0.09	4.48 r
42	**gentbk1.pcpi_mul/mul_2209/U295/CO (FADDX1_LVT)**	0.09	4.57 r
43	**gentbk1.pcpi_mul/mul_2209/U294/CO (FADDX1_LVT)**	0.09	4.67 r
44	**gentbk1.pcpi_mul/mul_2209/U293/CO (FADDX1_LVT)**	0.09	4.76 r
45	**gentbk1.pcpi_mul/mul_2209/U292/CO (FADDX1_LVT)**	0.09	4.86 r
46	**gentbk1.pcpi_mul/mul_2209/U291/CO (FADDX1_LVT)**	0.09	4.95 r
47	**gentbk1.pcpi_mul/mul_2209/U290/CO (FADDX1_LVT)**	0.09	5.05 r
48	**gentbk1.pcpi_mul/mul_2209/U289/CO (FADDX1_LVT)**	0.09	5.14 r
49	**gentbk1.pcpi_mul/mul_2209/U235/Y (NAND2X0_LVT)**	0.03	5.18 f
50	**gentbk1.pcpi_mul/mul_2209/U233/Y (NAND2X0_LVT)**	0.05	5.23 r
51	**gentbk1.pcpi_mul/mul_2209/U236/Y (NAND2X0_LVT)**	0.03	5.26 f
52	**gentbk1.pcpi_mul/mul_2209/U230/Y (NAND2X0_LVT)**	0.05	5.31 r
53	**gentbk1.pcpi_mul/mul_2209/U288/CO (FADDX1_LVT)**	0.10	5.41 r
54	**gentbk1.pcpi_mul/mul_2209/U287/CO (FADDX1_LVT)**	0.09	5.50 r
55	**gentbk1.pcpi_mul/mul_2209/U286/CO (FADDX1_LVT)**	0.09	5.60 r
56	**gentbk1.pcpi_mul/mul_2209/U285/CO (FADDX1_LVT)**	0.09	5.69 r
57	**gentbk1.pcpi_mul/mul_2209/U282/CO (FADDX1_LVT)**	0.09	5.79 r
58	**gentbk1.pcpi_mul/mul_2209/U231/Y (NAND2X0_LVT)**	0.03	5.82 f
59	**gentbk1.pcpi_mul/mul_2209/U246/Y (NAND2X0_LVT)**	0.05	5.87 r
60	**gentbk1.pcpi_mul/mul_2209/U280/CO (FADDX1_LVT)**	0.10	5.97 r
61	**gentbk1.pcpi_mul/mul_2209/U279/CO (FADDX1_LVT)**	0.09	6.06 r
62	**gentbk1.pcpi_mul/mul_2209/U278/CO (FADDX1_LVT)**	0.09	6.16 r
63	**gentbk1.pcpi_mul/mul_2209/U277/CO (FADDX1_LVT)**	0.09	6.25 r
64	**gentbk1.pcpi_mul/mul_2209/U276/CO (FADDX1_LVT)**	0.09	6.35 r
65	**gentbk1.pcpi_mul/mul_2209/U275/CO (FADDX1_LVT)**	0.09	6.44 r
66	**gentbk1.pcpi_mul/mul_2209/U274/CO (FADDX1_LVT)**	0.09	6.54 r
67	**gentbk1.pcpi_mul/mul_2209/U273/CO (FADDX1_LVT)**	0.09	6.63 r
68	**gentbk1.pcpi_mul/mul_2209/U272/CO (FADDX1_LVT)**	0.09	6.73 r
69	**gentbk1.pcpi_mul/mul_2209/U271/CO (FADDX1_LVT)**	0.09	6.82 r
70	**gentbk1.pcpi_mul/mul_2209/U270/CO (FADDX1_LVT)**	0.09	6.91 r
71	**gentbk1.pcpi_mul/mul_2209/U269/CO (FADDX1_LVT)**	0.09	7.01 r
72	**gentbk1.pcpi_mul/mul_2209/U268/CO (FADDX1_LVT)**	0.09	7.10 r
73	**gentbk1.pcpi_mul/mul_2209/U267/CO (FADDX1_LVT)**	0.09	7.20 r
74	**gentbk1.pcpi_mul/mul_2209/U266/CO (FADDX1_LVT)**	0.09	7.29 r
75	**gentbk1.pcpi_mul/mul_2209/U215/Y (XOR3X2_LVT)**	0.06	7.35 r
76	**gentbk1.pcpi_mul/mul_2209/product[63] (picorv32_pcpi_fast_mul_DW_mult_tc_0)**	0.00	7.35 r
77	**gentbk1.pcpi_mul/mul_reg[63]/D (DFFX1_LVT)**	0.01	7.36 r
78	**data arrival time**		7.36
79	**clock clk (rise edge)**	6.00	6.00
80	**clock network delay (ideal)**	1.00	7.00
81	**clock uncertainty**	-1.00	6.00
82	**gentbk1.pcpi_mul/mul_reg[63]/CLK (DFFX1_LVT)**	0.00	6.00 r
83	**library setup time**	-0.07	5.93
84	**data required time**		5.93
86	**data required time**		**5.93**
87	**data arrival time**		**-7.36**
	slack (VIOLATED)		**-1.43**

Report_timing - 1 (Compile)-1

S.No	
1	**Startpoint:** instr_lw_reg (rising edge-triggered flip-flop clocked by clk)
2	**Endpoint:** is_lbu_lhu_lw_reg (rising edge-triggered flip-flop clocked by clk)
3	**Path Group:** Clk
4	**Path Type:** Min

Report_timing-2 (Compile)-1

Des/Clust/Port	Wire Load Model	Library
picorv32a	**35000**	**saed32rvt_ttftp7ltvs40c**

Report_timing -3 (Compile)-1

S.No	Point	Incr	Path
1	**clock clk (rise edge)**	0.00	0.00
2	**clock network delay (ideal)**	0.00	0.00
3	**instr_lw_reg/CLK (DFFX1_LVT)**	0.00 #	0.00 r
4	**instr_lw_reg/QN (DFFX1_LVT)**	0.06	0.06 f
5	**U3354/Y (NAND2X0_LVT)**	0.04	0.10 r
6	**is_lbu_lhu_lw_reg/D (DFFX1_LVT)**	0.01	0.11 r
7	**data arrival time**		0.11
8	**clock clk (rise edge)**	0.00	0.00
9	**clock network delay (ideal)**	0.00	0.00
10	**is_lbu_lhu_lw_reg/CLK (DFFX1_LVT)**	0.00	0.00 r
11	**library hold time**	-0.02	-0.02
	data required time		-0.02
12	**data required time**		-0.02
13	**data arrival time**		-0.11
	slack (MET)		**0.13**

Report- Threshold Voltage Group-compile

S.No	Vth Group Name->	saed32svt_ll	saed32svt_svt	Total
2	**All Cells**	7619 (89.92%)	854 (10.08%)	8473 (100.00%)
3	**Blackbox Cells**	0 (0.00%)	0 (0.00%)	0 (0.00%)
4	**Non-Blackbox cells**	7619 (89.92%)	854 (10.08%)	8473 (100.00%)
5	**All Cells Area**	26258.41	3348.60 (11.31%)	29607.01
6	**Blackbox Cells area**	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)
7	**Non-Blackbox cells area**	26258.41 (88.69%)	3348.60 (11.31%)	29607.01 (100.00%)
8	**All Cells leakage**	3.114nW (99.80%)	6.150nW (0.20%)	3.120nW (100.00%)
9	**Blackbox Cells leakage**	0.000pW (0.00%)	0.000pW (0.00%)	0.000pW (0.00%)
10	**Non-Blackbox cells leakage**	3.114nW (99.80%)	6.150nW (0.20%)	3.120nW (100.00%)

Report hierarchy (compile)

picorv32a	r
AND2X1_LVT	saed32rvt_ttftp7ltvs40c
AND2X1_RVT	saed32rvt_ttftp7ltvs40c
AND2X2_LVT	saed32rvt_ttftp7ltvs40c
AND3X1_LVT	saed32rvt_ttftp7ltvs40c
AND3X2_LVT	saed32rvt_ttftp7ltvs40c
AND4X1_LVT	saed32rvt_ttftp7ltvs40c
AND4X2_LVT	saed32rvt_ttftp7ltvs40c
AO21X1_LVT	saed32rvt_ttftp7ltvs40c
AO22X1_LVT	saed32rvt_ttftp7ltvs40c
AO22X1_RVT	saed32rvt_ttftp7ltvs40c
AO221X1_LVT	saed32rvt_ttftp7ltvs40c
AO221X1_RVT	saed32rvt_ttftp7ltvs40c
AO221X2_LVT	saed32rvt_ttftp7ltvs40c
AO222X1_LVT	saed32rvt_ttftp7ltvs40c
AO31X1_LVT	saed32rvt_ttftp7ltvs40c
AOI22X1_LVT	saed32rvt_ttftp7ltvs40c
AOI221X1_LVT	saed32rvt_ttftp7ltvs40c
AOI222X1_LVT	saed32rvt_ttftp7ltvs40c
DFFX1_LVT	saed32rvt_ttftp7ltvs40c
DFFX1_RVT	saed32rvt_ttftp7ltvs40c
DFFX2_LVT	saed32rvt_ttftp7ltvs40c
INV0_LVT	saed32rvt_ttftp7ltvs40c
INV0_RVT	saed32rvt_ttftp7ltvs40c
INV1_LVT	saed32rvt_ttftp7ltvs40c
INV2_LVT	saed32rvt_ttftp7ltvs40c
INV4_LVT	saed32rvt_ttftp7ltvs40c
LATCHX1_LVT	saed32rvt_ttftp7ltvs40c
LATCHX1_RVT	saed32rvt_ttftp7ltvs40c
MUX21X1_LVT	saed32rvt_ttftp7ltvs40c
MUX21X1_RVT	saed32rvt_ttftp7ltvs40c
NAND2X0_LVT	saed32rvt_ttftp7ltvs40c
NAND2X0_RVT	saed32rvt_ttftp7ltvs40c
NAND2X2_LVT	saed32rvt_ttftp7ltvs40c
NAND3X0_LVT	saed32rvt_ttftp7ltvs40c
NAND3X0_RVT	saed32rvt_ttftp7ltvs40c
NAND4X0_LVT	saed32rvt_ttftp7ltvs40c
NAND4X0_RVT	saed32rvt_ttftp7ltvs40c
NBUFFX2_LVT	saed32rvt_ttftp7ltvs40c
NBUFFX2_RVT	saed32rvt_ttftp7ltvs40c
NOR0_LVT	saed32rvt_ttftp7ltvs40c
NOR0_RVT	saed32rvt_ttftp7ltvs40c
NOR4X1_LVT	saed32rvt_ttftp7ltvs40c
OA21X1_LVT	saed32rvt_ttftp7ltvs40c
**OA22X

Report-cell

S.No	Reference	Library	Unit Area	Cell Count	Total Area	Attributes
1	AND2X1_LVT	sand32vt_tt0p78vn40c	2.033152	9	18.298369	
2	AND2X1_RVT	sand32vt_tt0p78vn40c	2.033152	762	1549.261903	
3	AND2X2_LVT	sand32vt_tt0p78vn40c	2.287296	2	4.574592	
4	AND2X2_RVT	sand32vt_tt0p78vn40c	2.287296	22	50.320313	
5	AND2X4_LVT	sand32vt_tt0p78vn40c	2.795584	1	2.795584	
6	AND2X4_RVT	sand32vt_tt0p78vn40c	2.795584	1	2.795584	
7	AND3X1_RVT	sand32vt_tt0p78vn40c	2.287296	67	153.248836	
8	AND3X2_RVT	sand32vt_tt0p78vn40c	2.541440	2	5.082880	
9	AND4X1_LVT	sand32vt_tt0p78vn40c	2.541440	1	2.541440	
10	AND4X1_RVT	sand32vt_tt0p78vn40c	2.541440	11	27.955840	
11	AO21X1_LVT	sand32vt_tt0p78vn40c	2.541440	19	48.287360	
12	AO21X1_RVT	sand32vt_tt0p78vn40c	2.541440	179	454.917762	
13	AO21X2_RVT	sand32vt_tt0p78vn40c	2.795584	1	2.795584	
14	AO22X1_LVT	sand32vt_tt0p78vn40c	2.541440	17	43.204480	
15	AO22X1_RVT	sand32vt_tt0p78vn40c	2.541440	955	2427.075210	
16	AO22X2_RVT	sand32vt_tt0p78vn40c	3.049728	9	27.447551	
17	AO22X4_RVT	sand32vt_tt0p78vn40c	3.303872	5	16.519361	
18	AOI21X1_LVT	sand32vt_tt0p78vn40c	3.049728	2	6.099456	
19	AOI21X1_RVT	sand32vt_tt0p78vn40c	3.049728	64	195.182587	
20	AOI22X1_LVT	sand32vt_tt0p78vn40c	3.049728	1	3.049728	
21	AOI22X1_RVT	sand32vt_tt0p78vn40c	3.049728	302	921.017831	
22	AOI222X1_RVT	sand32vt_tt0p78vn40c	3.812160	1	3.812160	
23	DFFSSRX1_RVT	sand32vt_tt0p78vn40c	7.116032	1	7.116032	n
24	DFFX1_LVT	sand32vt_tt0p78vn40c	6.607744	240	1585.858612	n
25	DFFX1_RVT	sand32vt_tt0p78vn40c	6.607744	1206	7968.939526	n
26	DFFX2_RVT	sand32vt_tt0p78vn40c	7.116032	2	14.232064	n
27	FADD1X1_LVT	sand32vt_tt0p78vn40c	4.828736	6	28.972415	n
28	FADD1X1_RVT	sand32vt_tt0p78vn40c	4.828736	2	9.657472	n
29	HADD1X1_RVT	sand32vt_tt0p78vn40c	3.303872	9	29.734849	n
30	HUFFX16_RVT	sand32vt_tt0p78vn40c	6.607744	2	13.215488	
31	INVX0_LVT	sand32vt_tt0p78vn40c	1.270720	33	41.933760	
32	INVX0_RVT	sand32vt_tt0p78vn40c	1.270720	699	888.233284	
33	INVX1_LVT	sand32vt_tt0p78vn40c	1.270720	9	11.436480	
34	INVX1_RVT	sand32vt_tt0p78vn40c	1.270720	35	44.475200	
35	INVX2_LVT	sand32vt_tt0p78vn40c	1.524864	4	6.099456	
36	INVX2_RVT	sand32vt_tt0p78vn40c	1.524864	22	33.547007	
37	INVX4_LVT	sand32vt_tt0p78vn40c	2.033152	1	2.033152	
38	INVX4_RVT	sand32vt_tt0p78vn40c	2.033152	4	8.132608	
39	INVX8_RVT	sand32vt_tt0p78vn40c	3.049728	1	3.049728	
40	LATCHX1_RVT	sand32vt_tt0p78vn40c	5.082880	68	345.635841	n
41	MUX21X1_RVT	sand32vt_tt0p78vn40c	3.303872	57	188.320710	
42	NAND2X0_LVT	sand32vt_tt0p78vn40c	1.524864	15	22.872959	
43	NAND2X0_RVT	sand32vt_tt0p78vn40c	1.524864	453	690.763373	
44	NAND2X4_LVT	sand32vt_tt0p78vn40c	3.303872	1	3.303872	
45	NAND3X0_LVT	sand32vt_tt0p78vn40c	1.779008	4	7.116032	
46	NAND3X0_RVT	sand32vt_tt0p78vn40c	1.779008	116	206.364932	
47	NAND3X1_RVT	sand32vt_tt0p78vn40c	2.795584	2	5.591168	
48	NAND3X2_RVT	sand32vt_tt0p78vn40c	3.049728	3	9.149184	
49	NAND4X0_LVT	sand32vt_tt0p78vn40c	2.033152	12	24.397825	
50	NAND4X0_RVT	sand32vt_tt0p78vn40c	2.033152	107	217.547275	
51	NAND4X1_LVT	sand32vt_tt0p78vn40c	3.049728	1	3.049728	
52	NHUFFX2_LVT	sand32vt_tt0p78vn40c	2.033152	2	4.066304	
53	NHUFFX2_RVT	sand32vt_tt0p78vn40c	2.033152	37	75.226628	
54	NHUFFX4_RVT	sand32vt_tt0p78vn40c	2.541440	1	2.541440	
55	NOR2X0_LVT	sand32vt_tt0p78vn40c	2.541440	17	43.204480	
56	NOR2X0_RVT	sand32vt_tt0p78vn40c	2.541440	450	1143.648005	
57	NOR2X2_RVT	sand32vt_tt0p78vn40c	2.795584	8	22.364672	
58	NOR2X4_RVT	sand32vt_tt0p78vn40c	3.303872	1	3.303872	
59	NOR3X0_LVT	sand32vt_tt0p78vn40c	2.795584	3	8.386752	
60	NOR3X0_RVT	sand32vt_tt0p78vn40c	2.795584	72	201.282045	
61	NOR3X1_RVT	sand32vt_tt0p78vn40c	3.049728	3	9.149184	
62	NOR4X0_RVT	sand32vt_tt0p78vn40c	3.049728	36	109.790205	
63	NOR4X1_LVT	sand32vt_tt0p78vn40c	3.049728	3	9.149184	
64	OA21X1_LVT	sand32vt_tt0p78vn40c	2.541440	3	7.624320	
65	OA21X1_RVT	sand32vt_tt0p78vn40c	2.541440	109	277.016961	
66	OA22X1_LVT	sand32vt_tt0p78vn40c	2.541440	1	2.541440	
67	OA22X1_RVT	sand32vt_tt0p78vn40c	2.541440	205	520.995202	
68	OA22X2_RVT	sand32vt_tt0p78vn40c	2.795584	1	2.795584	
69	OA22X4_RVT	sand32vt_tt0p78vn40c	3.049728	1	3.049728	
70	OA222X1_RVT	sand32vt_tt0p78vn40c	3.303872	3	9.911616	
71	OAI21X1_LVT	sand32vt_tt0p78vn40c	3.049728	7	21.348095	
72	OAI21X1_RVT	sand32vt_tt0p78vn40c	3.049728	215	655.691302	
73	OAI21X2_RVT	sand32vt_tt0p78vn40c	3.303872	1	3.303872	
74	OAI22X1_LVT	sand32vt_tt0p78vn40c	3.049728	2	6.099456	
75	OAI22X1_RVT	sand32vt_tt0p78vn40c	3.049728	181	552.000753	
76	OAI22X2_RVT	sand32vt_tt0p78vn40c	3.812160	12	45.745920	
77	OR2X1_LVT	sand32vt_tt0p78vn40c	2.033152	15	30.49728	
78	OR2X1_RVT	sand32vt_tt0p78vn40c	2.033152	516	1049.106485	
79	OR2X2_RVT	sand32vt_tt0p78vn40c	2.287296	5	11.436480	
80	OR2X4_RVT	sand32vt_tt0p78vn40c	2.795584	3	8.386752	
81	OR3X1_LVT	sand32vt_tt0p78vn40c	2.287296	4	9.149184	
82	OR3X1_RVT	sand32vt_tt0p78vn40c	2.541440	84	213.480961	
83	OR3X2_LVT	sand32vt_tt0p78vn40c	2.541440	1	2.541440	
84	OR3X2_RVT	sand32vt_tt0p78vn40c	2.541440	8	20.331520	
85	OR3X4_RVT	sand32vt_tt0p78vn40c	4.066304	1	4.066304	
86	OR4X1_LVT	sand32vt_tt0p78vn40c	3.558016	1	3.558016	
87	OR4X1_RVT	sand32vt_tt0p78vn40c	3.558016	37	131.646594	
88	OR4X2_RVT	sand32vt_tt0p78vn40c	3.812160	19	72.431040	
89	XNOR2X1_LVT	sand32vt_tt0p78vn40c	4.320448	3	12.961344	
90	XNOR2X1_RVT	sand32vt_tt0p78vn40c	4.320448	157	678.310324	
91	XNOR3X1_LVT	sand32vt_tt0p78vn40c	6.099456	2	12.198912	
92	XNOR3X1_RVT	sand32vt_tt0p78vn40c	6.099456	26	158.585856	
93	XOR2X1_LVT	sand32vt_tt0p78vn40c	4.320448	2	8.640896	
94	XOR2X1_RVT	sand32vt_tt0p78vn40c	4.320448	114	492.531063	
95	XOR3X1_LVT	sand32vt_tt0p78vn40c	7.116032	1	7.116032	
96	XOR3X1_RVT	sand32vt_tt0p78vn40c	7.116032	7	49.812225	
97	XOR3X2_LVT	sand32vt_tt0p78vn40c	7.370176	1	7.370176	
98	picorv32_pcpi_fast_mul		9691.781351	1	9691.781351	h, n
	Total top module cells			**7930**		
	Total no of cells			**10841**		

Report-area (Compile Ultra)

S.No	Attribute	Area	Count
1	**Number of ports:**		543
2	**Number of nets:**		11975
3	**Number of cells:**		10841
4	**Number of combinational cells:**		9185
5	**Number of sequential cells:**		1650
6	**Number of macros/black boxes:**		0
7	**Number of buf/inv:**		1224
8	**Number of references:**		98
9	**Combinational area:**	23998.309593	
10	**Buf/Inv area:**	1639.482953	
11	**Noncombinational area:**	10804.932503	
12	**Macro/Black Box area:**	0.000000	
13	**Net Interconnect area:**	11317.279926	
	Total cell area:	**34803.242096**	
	Total area:	**46120.522022**	

Report_power-1 (Compile Ultra)

S.No	Design	Wire Load Model	Library
1	picorv32a	35000	sand32vt_tt0p78vn4
2	picorv32_pcpi_fast_mul	16000	sand32vt_tt0p78vn4

Cell Internal Power = 1.1339 mW (99%)
Net Switching Power = 6.5549 uW (1%)

Total Dynamic Power = 1.1404 mW (100%)
Cell Leakage Power = 471.1416 uW

Report-power-2 (compile ultra)

S.No	Power Group	Internal Power	Switching Power	Total Power	Leakage Power	Percentage	Attributes
1	io_pad	0.0000	0.0000	0.0000	0.0000	0.00%	
2	memory	0.0000	0.0000	0.0000	0.0000	0.00%	
3	black_box	0.0000	0.0000	0.0000	0.0000	0.00%	
4	clock_network	1.1228E-03	0.0000	0.0000	1.1228E-03	69.67%	i
5	register	2.4453	1.0067	3.9237E-08	395.8158	24.56%	
6	sequential	0.4972	8.9704E-02	1.1333E-06	1.7203	0.11%	
7	combinational	8.0849	5.4584	7.7642E-07	91.1855	5.66%	
	Total (uW)	**1.1339E-03**	**6.5549**	**4.7114e-08 pW**	**1.6115E-03**		
		Total Dynamic Power	**1.1404 mW**				

Report_timing - 1 (Compile ultra)

S.No	
1	**Startpoint:**
2	**Endpoint:**
3	**Path Group:**
4	**Path Type:**

picorv32a	35000 sand32vt_tt0p78vn40c
picorv32_pcpi_fast_mul	16000 sand32vt_tt0p78vn40c

Report_timing-2 (Compile ultra)

Des/Clust/Port	Wire Load Model	Library
picorv32a	35000	sand32vt_tt0p78vn40c
picorv32_pcpi_fast_mul	16000	sand32vt_tt0p78vn40c

Report_timing -3 (Compile ultra)

S.No	Point	Incr	Path
1	**clock clk (rise edge)**	0.00	0.00
2	**clock network delay (Ideal)**	1.00	1.00
3	**genblk1.pcpi_mul/mul_reg[15]/CLK (DFFX1_RVT)**	0.00 #	1.00 r
4	**genblk1.pcpi_mul/mul_reg[15]/Q (DFFX1_RVT)**	0.30	1.30 f
5	**genblk1.pcpi_mul/U346/Y (NBUFFX4_LVT)**	0.08	1.38 f
6	**genblk1.pcpi_mul/U157/Y (NAND2X0_RVT)**	0.10	1.48 r
7	**genblk1.pcpi_mul/U285/Y (INVX0_RVT)**	0.07	1.53 f
8	**genblk1.pcpi_mul/U147/Y (NOR2X0_RVT)**	0.11	1.64 r
9	**genblk1.pcpi_mul/U167/Y (OA21X1_RVT)**	0.08	1.72 r
10	**genblk1.pcpi_mul/U246/Y (INVX0_RVT)**	0.05	1.77 f
11	**genblk1.pcpi_mul/U247/Y (AOI21X1_RVT)**	0.18	1.96 r
12	**genblk1.pcpi_mul/U306/Y (OA21X1_RVT)**	0.12	2.08 r
13	**genblk1.pcpi_mul/U75/Y (OA21X1_RVT)**	0.11	2.18 r
14	**genblk1.pcpi_mul/U175/Y (NOR2X1_RVT)**	0.17	2.36 f
15	**genblk1.pcpi_mul/U76/Y (NBUFFX2_RVT)**	0.10	2.45 f
16	**genblk1.pcpi_mul/U264/Y (OA21X1_RVT)**	0.18	2.63 r
17	**genblk1.pcpi_mul/U086/Y (NOR2X1_RVT)**	0.19	2.82 f
18	**genblk1.pcpi_mul/U110/S (FADD1X1_LVT)**	0.24	3.06 r
19	**genblk1.pcpi_mul/U215/S (FADD1X1_RVT)**	0.24	3.30 f
20	**genblk1.pcpi_mul/U275/Y (NOR2X1_RVT)**	0.29	3.59 f
21	**genblk1.pcpi_mul/U136/S (FADD1X1_LVT)**	0.24	3.83 r
22	**genblk1.pcpi_mul/U275/Y (NOR2X1_RVT)**	0.12	3.95 f
23	**genblk1.pcpi_mul/U333/Y (OA21X1_RVT)**	0.16	4.11 r
24	**genblk1.pcpi_mul/U223/Y (AOI21X1_RVT)**	0.12	4.23 f
25	**genblk1.pcpi_mul/U15/Y (OA21X1_RVT)**	0.12	4.35 r
26	**genblk1.pcpi_mul/U230/Y (AOI21X1_RVT)**	0.10	4.45 f
27	**genblk1.pcpi_mul/U233/Y (AOI21X1_RVT)**	0.08	4.53 r
28	**genblk1.pcpi_mul/U291/Y (AOI22X1_RVT)**	0.10	4.64 f
29	**genblk1.pcpi_mul/U201/CO (FADD1X1_LVT)**	0.09	4.73 r
30	**genblk1.pcpi_mul/U191/CO (FADD1X1_LVT)**	0.10	4.83 r
31	**genblk1.pcpi_mul/U400/CO (FADD1X1_LVT)**	0.10	4.93 r
32	**genblk1.pcpi_mul/U401/CO (FADD1X1_LVT)**	0.11	5.04 r
33	**genblk1.pcpi_mul/U406/CO (FADD1X1_LVT)**	0.12	5.16 r
34	**genblk1.pcpi_mul/U416/Y (AOI21X1_LVT)**	0.07	5.23 r
35	**genblk1.pcpi_mul/U404/CO (FADD1X1_LVT)**	0.11	5.34 r
36	**genblk1.pcpi_mul/U374/Y (AOI21X1_RVT)**	0.11	5.45 r
37	**genblk1.pcpi_mul/U287/CO (FADD1X1_RVT)**	0.18	5.63 r
38	**genblk1.pcpi_mul/U389/Y (AOI21X1_RVT)**	0.11	5.75 r
39	**genblk1.pcpi_mul/U371/Y (AOI21X1_RVT)**	0.10	5.85 r
40	**genblk1.pcpi_mul/U377/Y (NOR2X1_LVT)**	0.06	5.91 r
41	**genblk1.pcpi_mul/mul_reg[63]/D (DFFX1_RVT)**	0.01	5.92 r
42	**data arrival time**		5.92
43	**clock clk (rise edge)**	6.00	6.00
44	**clock network delay (Ideal)**	1.00	7.00
45	**clock uncertainty**	-1.00	6.00
46	**genblk1.pcpi_mul/mul_reg[63]/CLK (DFFX1_RVT)**	0.00	6.00 r
47	**library setup time**	-0.08	5.92
48	**data required time**		5.92
49	**data required time**		5.92
50	**data arrival time**		-5.92
	Slack (MET)		**0.00**

Report_timing - 1 (Compile)-1

S.No	
1	**Startpoint:**
2	**Endpoint:**
3	**Path Group:**
4	**Path Type:**

count_cycle_reg[6] (rising edge-triggered flip-flop clocked by clk)
count_cycle_reg[6] (rising edge-triggered flip-flop clocked by clk)
Clk
Min

Report_timing-2 (Compile)-1

Des/Clust/Port	Wire Load Model	Library
picorv32a	35000	sand32vt_tt0p78vn40c

Report_timing -3 (Compile)-1

S.No	Point	Incr	Path
1	**clock clk (rise edge)**	0.00	0.00
2	**clock network delay (Ideal)**	0.00	0.00
3	**count_cycle_reg[6]/CLK (DFFX1_LVT)**	0.00 #	0.00 r
4	**count_cycle_reg[6]/QN (DFFX1_LVT)**	0.06	0.06 f
5	**U9800/Y (AND2X1_RVT)**	0.07	0.13 f
6	**count_cycle_reg[5]/D (DFFX1_LVT)**	0.01	0.14 f
7	**data arrival time**		0.14
8	**clock clk (rise edge)**	0.00	0.00
9	**clock network delay (Ideal)**	0.00	0.00
10	**ix_0u_0u_ix_reg/CLK (DFFX1_LVT)**	0.00	0.00 r
11	**library hold time**	-0.02	-0.02
	data required time		-0.02
12	**data required time**		-0.02
13	**data arrival time**		-0.14
	slack (MET)		**0.16**

Report-Wire load Model (Compile ultra)-1

Wire load model:	**35000**
Location	**picorv32a (design)**
Resistance :	**0.001187**
Capacitance :	**0.00057**
Area :	**0.01**
Slope :	**140.36**

Report-Wire load Model -35000 (Compile ultra)

S.No	Fanout	Length	Points	Average cap	Standard Deviation	%SD
1	1	16.92				
2	2	39.03				
3	3	64.09				
4	4	92.31				
5	5	124.73				
6	6	161.17				
7	7	202.25				
8	8	248.42				
9	9	300.09				
10	10	357.68				
11	11	421.63				
12	12	492.37				
13	13	570.31				
14	14	655.89				
15	15	749.54				
16	16	851.67				
17	17	962.72				
18	18	1083.11				
19	19	1213.28				
20	20	1353.64				
Total Length						

Report-Wire load Model (Compile ultra)-2

Wire load model:	**16000**
Location	**picorv32_pcpi_fast_mul (design)**
Resistance :	**0.00128**
Capacitance :	**0.00056**
Area :	**0.01**
Slope :	**108.438**

Report-Wire load Model - 16000(Compile ultra)

S.No	Fanout	Length	Points	Average cap	Standard Deviation	%SD
1	1	15.10				
2	2	34.61				
3	3	56.42				
4	4	80.84				
5	5	108.20				
6	6	138.79				
7	7	172.92				
8	8	210.91				
9	9	253.07				
10	10	299.71				
11	11	351.13				
12	12	407.64				
13	13	469.57				
14	14	537.20				
15	15	610.87				
16	16	690.86				
17	17	777.51				
18	18	871.11				
19	19	971.97				
20	20	1080.41				
Total Length						

Report- Threshold Voltage Group-compile_ultra

S.No	Vth Group Name	sand32lvt_lvt	sand32vt_rvt	Total
2	**All Cells**	499 (4.61%)	10336 (95.39%)	10835
3	**Blackbox Cells**	0 (0.00%)	0 (0.00%)	0
4	**Non-Blackbox cells**	499 (4.61%)	10336 (95.39%)	10835
5	**All Cells Area**	2375.99 (6.83%)	32427.25 (93.17%)	34803.24
6	**Blackbox Cells area**	0 (0.00%)	0 (0.00%)	0
7	**Non-Blackbox cells area**	2375.99 (6.83%)	32427.25 (93.17%)	34803.24 (100.00%)
8	**All Cells leakage**	414.344nW (87.94%)	56.797nW (12.06%)	471.142nW (100.00%)
9	**Blackbox Cells leakage**	0 (0.00%)	0 (0.00%)	0
10	**Non-Blackbox cells leakage**	414.344nW (87.94%)	56.797nW (12.06%)	471.142nW (100.00%)

Global Skew Report(Compute ultra)

S.No		
1	**Clock Tree Name :**	"clk"
2	**Clock Period :**	6
3	**Clock Tree root pin :**	"clk"
4	**Number of Levels :**	1
5	**Number of Sinks :**	1582
6	**Number of CT Buffers :**	0
7	**Number of CTS added gates :**	0
8	**Number of Preexisting Gates :**	0
9	**Number of Preexisting Buf/Inv :**	0
10	**Total Number of Clock Cells :**	0
11	**Total Area of CT Buffers :**	0
12	**Total Area of CT cells :**	0
13	**Max Global Skew :**	0.00247
14	**Number of Max/Trans Violators :**	0
15	**Number of Max/Cap Violators :**	1
16	**Number of Max/Fanout Violators :**	0
1	**Operating Condition**	worst
2	**Clock global Skew**	0.002
3	**Longest path delay**	0.125
4	**Shortest path delay**	0.123

The longest path delay end pin: genblk1.pcpi_mul/mul2_reg[5]/CLK
The shortest path delay end pin: genblk1.pcpi_mul/mul1_reg[32]/CLK

Clock_tree Longest Path (Compute ultra)

S.No	Pin	Cap	Fanout	Trans	Incr	Arri
1	**clk**		0	1	0	0 r
2	**clk**	1060107.50f	1582	0	0	0 r
3	**genblk1.pcpi_mul/mul2_reg[5]/CLK**	1060107.50f	0	0	0.125	0.125 r
	Clock Delay					0.125

Clock_tree Shortest Path (Compute ultra)

S.No	Pin	Cap	Fanout	Trans	Incr	Arri
1	**clk**		0	1	0	0 r
2	**clk**	1060107.50f	1582	0	0	0 r
3	**genblk1.pcpi_mul/mul1_reg[32]/CLK**	1060107.50f	0	0	0.123	0.123 r
	Clock Delay					0.123

Report-qor (Compile_ultra)

S.No	Timing Path Group [none]
1	Levels of Logic:
2	Critical Path Length:
3	Critical Path Slack:
4	Critical Path Clk Period:
5	Total Negative Slack:
6	No. of Violating
7	Worst Hold
8	Total Hold Violation:
9	No. of Hold
	Cell Count
10	Hierarchical Cell
11	Hierarchical Port
12	Leaf Cell Count:
13	Buf/Inv Cell Count:
14	Buf Cell Count:
15	Inv Cell Count:
16	CT Buf/Inv Cell
17	Combinational Cell Count:
18	Sequential Cell Count:
19	Macro Count:
	Area
20	Combinational Area:
21	Noncombinational Area:
22	Buf/Inv Area:
23	Total Buffer Area:
24	Total Inverter Area:
25	Macro/Black Box
26	Net Area:
	Cell Area:
	Design Area:
	Design Rules
	Total Number of Nets:
	Nets With Violations:
	Max Trans
	Max Cap Violations:

36.00	
4.92	
0.00	
6.00	
0.00	
0.00	
0.00	
0.00	
0.00	
134	
10835	
1224	
59	
1165	
9185	
1650	
0	
23998.309593	
10804.932503	
1639.482953	
127.58	
1511.90	
0.000000	
11317.279926	
34803.242096	
46120.522022	
11858	
0	
0	
0	

Report hierarchy (compile_ultra)

picorv32a	
AND2X1_LVT	sand32vt_tt0p78vn40c
AND2X1_RVT	sand32vt_tt0p78vn40c
AND2X2_LVT	sand32vt_tt0p78vn40c
AND2X2_RVT	sand32vt_tt0p78vn40c
AND2X4_LVT	sand32vt_tt0p78vn40c
AND2X4_RVT	sand32vt_tt0p78vn40c
AND3X1_RVT	sand32vt_tt0p78vn40c
AND3X2_RVT	sand32vt_tt0p78vn40c
AND4X1_LVT	sand32vt_tt0p78vn40c
AND4X1_RVT	sand32vt_tt0p78vn40c
AO21X1_LVT	sand32vt_tt0p78vn40c
AO21X1_RVT	sand32vt_tt0p78vn40c
AO21X2_RVT	sand32vt_tt0p78vn40c
AO22X1_LVT	sand32vt_tt0p78vn40c
AO22X1_RVT	sand32vt_tt0p78vn40c
AO22X2_RVT	sand32vt_tt0p78vn40c
AO22X4_RVT	sand32vt_tt0p78vn40c
AOI21X1_LVT	sand32vt_tt0p78vn40c
AOI21X1_RVT	sand32vt_tt0p78vn40c
AOI22X1_LVT	sand32vt_tt0p78vn40c
AOI22X1_RVT	sand32vt_tt0p78vn40c
AOI222X1_RVT	sand32vt_tt0p78vn40c
DFFSSRX1_RVT	sand32vt_tt0p78vn40c
DFFX1_LVT	sand32vt_tt0p78vn40c
DFFX1_RVT	sand32vt_tt0p78vn40c
DFFX2_RVT	sand32vt_tt0p78vn40c
FADD1X1_LVT	sand32vt_tt0p78vn40c
FADD1X1_RVT	sand32vt_tt0p78vn40c
HADD1X1_RVT	sand32vt_tt0p78vn40c
HUFFX16_RVT	sand32vt_tt0p78vn40c
INVX0_LVT	sand32vt_tt0p78vn40c
INVX0_RVT	sand32vt_tt0p78vn40c
INVX1_LVT	sand32vt_tt0p78vn40c
INVX1_RVT	sand32vt_tt0p78vn40c
INVX2_LVT	sand32vt_tt0p78vn40c
INVX2_RVT	sand32vt_tt0p78vn40c
INVX4_LVT	sand32vt_tt0p78vn40c
INVX4_RVT	sand32vt_tt0p78vn40c
INVX8_RVT	sand32vt_tt0p78vn40c
LATCHX1_RVT	sand32vt_tt0p78vn40c
MUX21X1_RVT	sand32vt_tt0p78vn40c
NAND2X0_LVT	sand32vt_tt0p78vn40c
NAND2X0_RVT	sand32vt_tt0p78vn40c
NAND2X4_LVT	sand32vt_tt0p78vn40c
NAND3X0_LVT	sand32vt_tt0p78vn40c
NAND3X0_RVT	sand32vt_tt0p78vn40c
NAND3X1_RVT	sand32vt_tt0p78vn40c
NAND3X2_RVT	sand32vt_tt0p78vn40c
NAND4X0_LVT	sand32vt_tt0p78vn40c
NAND4X0_RVT	sand32vt_tt0p78vn40c
NAND4X1_LVT	sand32vt_tt0p78vn40c
NBUFFX2_LVT	sand32vt_tt0p78vn40c
NBUFFX2_RVT	sand32vt_tt0p78vn40c
NBUFFX4_RVT	sand32vt_tt0p78vn40c
NOR2X0_LVT	sand32vt_tt0p78vn40c
NOR2X0_RVT	sand32vt_tt0p78vn40c
NOR2X2_RVT	sand32vt_tt0p78vn40c
NOR2X4_RVT	sand32vt_tt0p78vn40c
NOR3X0_LVT	sand32vt_tt0p78vn40c
NOR3X0_RVT	sand32vt_tt0p78vn40c
NOR3X1_RVT	sand32vt_tt0p78vn40c
NOR4X0_RVT	sand32vt_tt0p78vn40c
NOR4X1_LVT	sand32vt_tt0p78vn40c
OA21X1_LVT	sand32vt_tt0p78vn40c
OA21X1_RVT	sand32vt_tt0p78vn40c
OA22X1_LVT	sand32vt_tt0p78vn40c
OA22X1_RVT	sand32vt_tt0p78vn40c
OA22X2_RVT	sand32vt_tt0p78vn40c
OA22X4_RVT	sand32vt_tt0p78vn40c
OA222X1_RVT	sand32vt_tt0p78vn40c
OAI21X1_LVT	sand32vt_tt0p78vn40c
OAI21X1_RVT	sand32vt_tt0p78vn40c
OAI21X2_RVT	sand32vt_tt0p78vn40c
OAI22X1_LVT	sand32vt_tt0p78vn40c
OAI22X1_RVT	sand32vt_tt0p78vn40c
OAI22X2_RVT	sand32vt_tt0p78vn40c
OR2X1_LVT	sand32vt_tt0p78vn40c
OR2X1_RVT	sand32vt_tt0p78vn40c
OR2X2_RVT	sand32vt_tt0p78vn40c
OR2X4_RVT	sand32vt_tt0p78vn40c
OR3X1_LVT	sand32vt_tt0p78vn40c
OR3X1_RVT	sand32vt_tt0p78vn40c
OR3X2_LVT	sand32vt_tt0p78vn40c
OR3X2_RVT	sand32vt_tt0p78vn40c
OR3X4_RVT	sand32vt_tt0p78vn40c
OR4X1_LVT	sand32vt_tt0p78vn40c
OR4X1_RVT	sand32vt_tt0p78vn40c
OR4X2_RVT	sand32vt_tt0p78vn40c
XNOR2X1_LVT	sand32vt_tt0p78vn40c
XNOR2X1_RVT	sand32vt_tt0p78vn40c
XNOR3X1_LVT	sand32vt_tt0p78vn40c
XNOR3X1_RVT	sand32vt_tt0p78vn40c
XOR2X1_LVT	sand

NLDM_Compile

S.No	Reference	Library	Unit Area	Cell Count	Total Area	Attributes
1	AND2X1_LVT	saed32lvt_tt0p78vn40c	2.033152	253	514.387482	
2	AND2X1_RVT	saed32rvt_tt0p78vn40c	2.033152	2	4.066304	
3	AND2X2_LVT	saed32lvt_tt0p78vn40c	2.287296	1	2.287296	
4	AND3X1_LVT	saed32lvt_tt0p78vn40c	2.287296	38	86.917250	
5	AND3X2_LVT	saed32lvt_tt0p78vn40c	2.541440	1	2.541440	
6	AND4X1_LVT	saed32lvt_tt0p78vn40c	2.541440	19	48.287360	
7	AND4X1_RVT	saed32rvt_tt0p78vn40c	2.541440	1	2.541440	
8	AND4X2_LVT	saed32lvt_tt0p78vn40c	3.812160	1	3.812160	
9	AO21X1_LVT	saed32lvt_tt0p78vn40c	2.541440	103	261.768321	
10	AO22X1_LVT	saed32lvt_tt0p78vn40c	2.541440	1116	2836.247051	
11	AO22X1_RVT	saed32rvt_tt0p78vn40c	2.541440	3	7.624320	
12	AO221X1_LVT	saed32lvt_tt0p78vn40c	3.049728	171	521.503474	
13	AO221X1_RVT	saed32rvt_tt0p78vn40c	3.049728	30	91.491838	
14	AO221X2_LVT	saed32lvt_tt0p78vn40c	3.303872	32	105.723907	
15	AO222X1_LVT	saed32lvt_tt0p78vn40c	3.303872	214	707.028631	
16	AO222X1_RVT	saed32rvt_tt0p78vn40c	3.303872	1	3.303872	
17	AOI21X1_LVT	saed32lvt_tt0p78vn40c	3.049728	5	15.248640	
18	AOI221X1_LVT	saed32lvt_tt0p78vn40c	3.558016	24	85.392385	
19	AOI222X1_LVT	saed32lvt_tt0p78vn40c	3.812160	24	91.491840	
20	DFFX1_LVT	saed32lvt_tt0p78vn40c	6.607744	1229	8120.917643	n
21	DFFX1_RVT	saed32rvt_tt0p78vn40c	6.607744	205	1354.587564	n
22	DFFX2_LVT	saed32lvt_tt0p78vn40c	7.116032	32	227.713028	n
23	INVX0_LVT	saed32lvt_tt0p78vn40c	1.270720	7	8.895040	
24	INVX0_RVT	saed32rvt_tt0p78vn40c	1.270720	3	3.812160	
25	INVX1_LVT	saed32lvt_tt0p78vn40c	1.270720	595	756.078403	
26	INVX2_LVT	saed32lvt_tt0p78vn40c	1.524864	9	13.723776	
27	INVX4_LVT	saed32lvt_tt0p78vn40c	2.033152	8	16.265217	
28	LATCHX1_LVT	saed32lvt_tt0p78vn40c	5.082880	63	320.221441	n
29	LATCHX1_RVT	saed32rvt_tt0p78vn40c	5.082880	5	25.414400	n
30	MUX21X1_LVT	saed32lvt_tt0p78vn40c	3.303872	59	194.928454	
31	MUX21X1_RVT	saed32rvt_tt0p78vn40c	3.303872	230	759.890585	
32	NAND2X0_LVT	saed32lvt_tt0p78vn40c	1.524864	135	205.856634	
33	NAND2X0_RVT	saed32rvt_tt0p78vn40c	1.524864	2	3.049728	
34	NAND2X2_LVT	saed32lvt_tt0p78vn40c	2.795584	1	2.795584	
35	NAND3X0_LVT	saed32lvt_tt0p78vn40c	1.779008	66	117.414530	
36	NAND3X0_RVT	saed32rvt_tt0p78vn40c	1.779008	3	5.337024	
37	NAND4X0_LVT	saed32lvt_tt0p78vn40c	2.033152	115	233.812492	
38	NBUFFX2_LVT	saed32lvt_tt0p78vn40c	2.033152	15	30.497282	
39	NBUFFX2_RVT	saed32rvt_tt0p78vn40c	2.033152	2	4.066304	
40	NOR2X0_LVT	saed32lvt_tt0p78vn40c	2.541440	7	17.790080	
41	NOR3X0_LVT	saed32lvt_tt0p78vn40c	2.795584	5	13.977920	
42	NOR4X1_LVT	saed32lvt_tt0p78vn40c	3.049728	12	36.596735	
43	OA21X1_LVT	saed32lvt_tt0p78vn40c	2.541440	13	33.038720	
44	OA22X1_LVT	saed32lvt_tt0p78vn40c	2.541440	43	109.281920	
45	OA221X1_LVT	saed32lvt_tt0p78vn40c	3.049728	167	509.304562	
46	OA222X1_LVT	saed32lvt_tt0p78vn40c	3.303872	160	528.619537	
47	OAI21X1_LVT	saed32lvt_tt0p78vn40c	3.049728	9	27.447551	
48	OAI21X1_RVT	saed32rvt_tt0p78vn40c	3.049728	31	94.541565	
49	OAI22X1_LVT	saed32lvt_tt0p78vn40c	3.049728	17	51.845375	
50	OAI221X1_RVT	saed32rvt_tt0p78vn40c	3.558016	3	10.674048	
51	OAI222X1_RVT	saed32rvt_tt0p78vn40c	3.812160	1	3.812160	
52	OR2X1_LVT	saed32lvt_tt0p78vn40c	2.033152	16	32.530434	
53	OR3X1_LVT	saed32lvt_tt0p78vn40c	2.287296	4	9.149184	
54	OR4X1_LVT	saed32lvt_tt0p78vn40c	3.558016	15	53.370241	
55	XNOR2X1_LVT	saed32lvt_tt0p78vn40c	4.320448	1	4.320448	
56	XNOR2X1_RVT	saed32rvt_tt0p78vn40c	4.320448	1	4.320448	
57	XOR2X1_RVT	saed32rvt_tt0p78vn40c	4.320448	1	4.320448	
58	picorv32_pcpi_fast_mul		7840.596464	1	7840.596464	h, n
59	picorv32a_DW01_add_0		156.044411	1	156.044411	h
60	picorv32a_DW01_add_1		151.215675	1	151.215675	h
61	picorv32a_DW01_add_2		182.983680	1	182.983680	h
62	picorv32a_DW01_add_3		192.895296	1	192.895296	h
63	picorv32a_DW01_add_4		156.044411	1	156.044411	h
64	picorv32a_DW01_add_5		156.044411	1	156.044411	h
65	picorv32a_DW01_add_6		188.066561	1	188.066561	h
66	picorv32a_DW01_cmp6_0		199.248897	1	199.248897	h
67	picorv32a_DW01_dec_0_DW01_d		181.458819	1	181.458819	h
68	picorv32a_DW01_inc_0		210.431239	1	210.431239	h
69	picorv32a_DW01_inc_1		210.431239	1	210.431239	h
70	picorv32a_DW01_sub_0		196.707451	1	196.707451	h
71	picorv32a_DW_cmp_0		242.961665	1	242.961665	h
	Total top module cells			**5343**		
	Total no of cells			**8598**		

NLDM_COMPILE_ULTRA

S.No	Reference	Library	Unit Area	Cell Count	Total Area	Attributes
1	AND2X1_LVT	saed32lvt_tt0p78vn40c	2.033152	9	18.298369	
2	AND2X1_RVT	saed32rvt_tt0p78vn40c	2.033152	762	1549.261903	
3	AND2X2_LVT	saed32lvt_tt0p78vn40c	2.287296	2	4.574592	
4	AND2X2_RVT	saed32rvt_tt0p78vn40c	2.287296	22	50.320513	
5	AND2X4_LVT	saed32lvt_tt0p78vn40c	2.795584	1	2.795584	
6	AND2X4_RVT	saed32rvt_tt0p78vn40c	2.795584	1	2.795584	
7	AND3X1_RVT	saed32rvt_tt0p78vn40c	2.287296	67	153.248836	
8	AND3X2_RVT	saed32rvt_tt0p78vn40c	2.541440	2	5.082880	
9	AND4X1_LVT	saed32lvt_tt0p78vn40c	2.541440	1	2.541440	
10	AND4X1_RVT	saed32rvt_tt0p78vn40c	2.541440	11	27.955840	
11	AO21X1_LVT	saed32lvt_tt0p78vn40c	2.541440	19	48.287360	
12	AO21X1_RVT	saed32rvt_tt0p78vn40c	2.541440	179	454.917762	
13	AO21X2_RVT	saed32rvt_tt0p78vn40c	2.795584	1	2.795584	
14	AO22X1_LVT	saed32lvt_tt0p78vn40c	2.541440	17	43.204480	
15	AO22X1_RVT	saed32rvt_tt0p78vn40c	2.541440	955	2427.075210	
16	AO221X1_RVT	saed32rvt_tt0p78vn40c	3.049728	9	27.447551	
17	AO222X1_RVT	saed32rvt_tt0p78vn40c	3.303872	5	16.519361	
18	AOI21X1_LVT	saed32lvt_tt0p78vn40c	3.049728	2	6.099456	
19	AOI21X1_RVT	saed32rvt_tt0p78vn40c	3.049728	64	195.182587	
20	AOI22X1_LVT	saed32lvt_tt0p78vn40c	3.049728	1	3.049728	
21	AOI22X1_RVT	saed32rvt_tt0p78vn40c	3.049728	302	921.017831	
22	AOI222X1_RVT	saed32rvt_tt0p78vn40c	3.812160	1	3.812160	
23	DFFSSRX1_RVT	saed32rvt_tt0p78vn40c	7.116032	1	7.116032	n
24	DFFX1_LVT	saed32lvt_tt0p78vn40c	6.607744	240	1585.858612	n
25	DFFX1_RVT	saed32rvt_tt0p78vn40c	6.607744	1206	7968.939526	n
26	DFFX2_RVT	saed32rvt_tt0p78vn40c	7.116032	2	14.232064	n
27	FADDX1_LVT	saed32lvt_tt0p78vn40c	4.828736	6	28.972415	r
28	FADDX1_RVT	saed32rvt_tt0p78vn40c	4.828736	2	9.657472	r
29	HADDX1_RVT	saed32rvt_tt0p78vn40c	3.303872	9	29.734849	r
30	IBUFFX16_RVT	saed32rvt_tt0p78vn40c	6.607744	2	13.215488	
31	INVX0_LVT	saed32lvt_tt0p78vn40c	1.270720	33	41.933760	
32	INVX0_RVT	saed32rvt_tt0p78vn40c	1.270720	699	888.233284	
33	INVX1_LVT	saed32lvt_tt0p78vn40c	1.270720	9	11.436480	
34	INVX1_RVT	saed32rvt_tt0p78vn40c	1.270720	35	44.475200	
35	INVX2_LVT	saed32lvt_tt0p78vn40c	1.524864	4	6.099456	
36	INVX2_RVT	saed32rvt_tt0p78vn40c	1.524864	22	33.547007	
37	INVX4_LVT	saed32lvt_tt0p78vn40c	2.033152	1	2.033152	
38	INVX4_RVT	saed32rvt_tt0p78vn40c	2.033152	4	8.132608	
39	INVX8_RVT	saed32rvt_tt0p78vn40c	3.049728	1	3.049728	
40	LATCHX1_RVT	saed32rvt_tt0p78vn40c	5.082880	68	345.635841	n
41	MUX21X1_RVT	saed32rvt_tt0p78vn40c	3.303872	57	188.320710	
42	NAND2X0_LVT	saed32lvt_tt0p78vn40c	1.524864	15	22.872959	
43	NAND2X0_RVT	saed32rvt_tt0p78vn40c	1.524864	453	690.763373	
44	NAND2X4_LVT	saed32lvt_tt0p78vn40c	3.303872	1	3.303872	
45	NAND3X0_LVT	saed32lvt_tt0p78vn40c	1.779008	4	7.116032	
46	NAND3X0_RVT	saed32rvt_tt0p78vn40c	1.779008	116	206.364932	
47	NAND3X1_RVT	saed32rvt_tt0p78vn40c	2.795584	2	5.591168	
48	NAND3X2_RVT	saed32rvt_tt0p78vn40c	3.049728	3	9.149184	
49	NAND4X0_LVT	saed32lvt_tt0p78vn40c	2.033152	12	24.397825	
50	NAND4X0_RVT	saed32rvt_tt0p78vn40c	2.033152	107	217.547275	
51	NAND4X1_LVT	saed32lvt_tt0p78vn40c	3.049728	1	3.049728	
52	NBUFFX2_LVT	saed32lvt_tt0p78vn40c	2.033152	2	4.066304	
53	NBUFFX2_RVT	saed32rvt_tt0p78vn40c	2.033152	37	75.226628	
54	NBUFFX4_RVT	saed32rvt_tt0p78vn40c	2.541440	1	2.541440	
55	NOR2X0_LVT	saed32lvt_tt0p78vn40c	2.541440	17	43.204480	
56	NOR2X0_RVT	saed32rvt_tt0p78vn40c	2.541440	450	1143.648005	
57	NOR2X2_RVT	saed32rvt_tt0p78vn40c	2.795584	8	22.364672	
58	NOR2X4_RVT	saed32rvt_tt0p78vn40c	3.303872	1	3.303872	
59	NOR3X0_LVT	saed32lvt_tt0p78vn40c	2.795584	3	8.386752	
60	NOR3X0_RVT	saed32rvt_tt0p78vn40c	2.795584	72	201.282045	
61	NOR3X2_RVT	saed32rvt_tt0p78vn40c	3.049728	3	9.149184	
62	NOR4X0_RVT	saed32rvt_tt0p78vn40c	3.049728	36	109.790205	
63	NOR4X1_LVT	saed32lvt_tt0p78vn40c	3.049728	3	9.149184	
64	OA21X1_LVT	saed32lvt_tt0p78vn40c	2.541440	3	7.624320	
65	OA21X1_RVT	saed32rvt_tt0p78vn40c	2.541440	109	277.016961	
66	OA22X1_LVT	saed32lvt_tt0p78vn40c	2.541440	1	2.541440	
67	OA22X1_RVT	saed32rvt_tt0p78vn40c	2.541440	205	520.995202	
68	OA22X2_RVT	saed32rvt_tt0p78vn40c	2.795584	1	2.795584	
69	OA221X1_RVT	saed32rvt_tt0p78vn40c	3.049728	1	3.049728	
70	OA222X1_RVT	saed32rvt_tt0p78vn40c	3.303872	3	9.911616	
71	OAI21X1_LVT	saed32lvt_tt0p78vn40c	3.049728	7	21.348095	
72	OAI21X1_RVT	saed32rvt_tt0p78vn40c	3.049728	215	655.691502	
73	OAI21X2_RVT	saed32rvt_tt0p78vn40c	3.303872	1	3.303872	
74	OAI22X1_LVT	saed32lvt_tt0p78vn40c	3.049728	2	6.099456	
75	OAI22X1_RVT	saed32rvt_tt0p78vn40c	3.049728	181	552.000753	
76	OAI222X1_RVT	saed32rvt_tt0p78vn40c	3.812160	12	45.745920	
77	OR2X1_LVT	saed32lvt_tt0p78vn40c	2.033152	15	30.497282	
78	OR2X1_RVT	saed32rvt_tt0p78vn40c	2.033152	516	1049.106485	
79	OR2X2_RVT	saed32rvt_tt0p78vn40c	2.287296	5	11.436480	
80	OR2X4_RVT	saed32rvt_tt0p78vn40c	2.795584	3	8.386752	
81	OR3X1_LVT	saed32lvt_tt0p78vn40c	2.287296	4	9.149184	
82	OR3X1_RVT	saed32rvt_tt0p78vn40c	2.541440	84	213.480961	
83	OR3X2_LVT	saed32lvt_tt0p78vn40c	2.541440	1	2.541440	
84	OR3X2_RVT	saed32rvt_tt0p78vn40c	2.541440	8	20.331520	
85	OR3X4_RVT	saed32rvt_tt0p78vn40c	4.066304	1	4.066304	
86	OR4X1_LVT	saed32lvt_tt0p78vn40c	3.558016	1	3.558016	
87	OR4X1_RVT	saed32rvt_tt0p78vn40c	3.558016	37	131.646594	
88	OR4X2_RVT	saed32rvt_tt0p78vn40c	3.812160	19	72.431040	
89	XNOR2X1_LVT	saed32lvt_tt0p78vn40c	4.320448	3	12.961344	
90	XNOR2X1_RVT	saed32rvt_tt0p78vn40c	4.320448	157	678.310324	
91	XNOR3X1_LVT	saed32lvt_tt0p78vn40c	6.099456	2	12.198912	
92	XNOR3X1_RVT	saed32rvt_tt0p78vn40c	6.099456	26	158.585852	
93	XOR2X1_LVT	saed32lvt_tt0p78vn40c	4.320448	2	8.640896	
94	XOR2X1_RVT	saed32rvt_tt0p78vn40c	4.320448	114	492.531063	
95	XOR3X1_LVT	saed32lvt_tt0p78vn40c	7.116032	1	7.116032	
96	XOR3X1_RVT	saed32rvt_tt0p78vn40c	7.116032	7	49.812225	
97	XOR3X2_LVT	saed32lvt_tt0p78vn40c	7.370176	1	7.370176	
98	picorv32_pcpi_fast_mul		9691.781351	1	9691.781351	h, n
	Total top module cells			**7930**		
	Total no of cells			**10841**		

NLDM_COMPILE			
S.No	**Number of**	**Count**	**Area**
1	**Number of ports:**	1919	
2	**Number of nets:**	11618	
3	**Number of cells:**	8546	
4	**Number of combinational cells:**	6806	
5	**Number of sequential cells:**	1667	
6	**Number of macros/black boxes:**	0	
7	**Number of buf/inv:**	1143	
8	**Number of references:**	71	
9	**Combinational area:**		18679.329838
10	**Buf/Inv area:**		1580.521542
11	**Noncombinational area:**		10927.684057
12	**Macro/Black Box area:**		0.000000
13	**Net Interconnect area:**		8977.222396
	Total Cell Area		**29607.013896**
	Total Area		**38584.236291**

NLDM_COMPILE_ULTRA			
S.No		**Area**	**Count**
1	**Number of ports:**		543
2	**Number of nets:**		11975
3	**Number of cells:**		10841
4	**Number of combinational cells:**		9185
5	**Number of sequential cells:**		1650
6	**Number of macros/black boxes:**		0
7	**Number of buf/inv:**		1224
8	**Number of references:**		98
9	**Combinational area:**	23998.309593	
10	**Buf/Inv area:**	1639.482953	
11	**Noncombinational area:**	10804.932503	
12	**Macro/Black Box area:**	0.000000	
13	**Net Interconnect area:**	11317.279926	
	Total cell area:	**34803.242096**	
	Total area:	**46120.522022**	

NLDM_COMPILE-1			
S.No	**Design**	**Wire Load Model**	**Library**
1	**picorv32a**	35000	saed32rvt_tt0p78vn4
2	**picorv32_pcpi_fast_mul**	8000	saed32rvt_tt0p78vn4
3	**picorv32a_DW01_add_0**	ForQA	saed32rvt_tt0p78vn4
4	**picorv32a_DW01_add_1**	ForQA	saed32rvt_tt0p78vn4
5	**picorv32a_DW01_inc_0**	8000	saed32rvt_tt0p78vn4
6	**picorv32a_DW01_add_2**	ForQA	saed32rvt_tt0p78vn4
7	**picorv32a_DW01_inc_1**	8000	saed32rvt_tt0p78vn4
8	**picorv32a_DW01_add_3**	ForQA	saed32rvt_tt0p78vn4
9	**picorv32a_DW_cmp_0**	8000	saed32rvt_tt0p78vn4
10	**picorv32a_DW01_add_4**	ForQA	saed32rvt_tt0p78vn4
11	**picorv32a_DW01_sub_0**	ForQA	saed32rvt_tt0p78vn4
12	**picorv32a_DW01_add_5**	ForQA	saed32rvt_tt0p78vn4
13	**picorv32a_DW01_dec_0_DW01_dec_1**	ForQA	saed32rvt_tt0p78vn4
14	**picorv32a_DW01_cmp6_0**	ForQA	saed32rvt_tt0p78vn4
15	**picorv32_pcpi_fast_mul_DW_mult_tc_0**	8000	saed32rvt_tt0p78vn4
16	**picorv32a_DW01_add_6**	ForQA	saed32rvt_tt0p78vn4

Cell Internal Power = 1.7865 mW (100%)
Net Switching Power = 6.6951 uW (0%)

Total Dynamic Power = 1.7932 mW (100%)
Cell Leakage Power = 3.1202 mW

NLDM_COMPILE-2							
S.No	**Power Group**	**Internal Power**	**Switching Power**	**Leakage Power**	**Total Power**	**Percentage**	**Attributes**
1	**io_pad**	0.0000	0.0000	0.0000	0.0000	0.00%)	
2	**memory**	0.0000	0.0000	0.0000	0.0000	0.00%)	
3	**black_box**	0.0000	0.0000	0.0000	0.0000	0.00%)	
4	**clock_network**	1.7812E+03	0.0000	0.0000	1.7812E+03	36.25%)	I
5	**register**	-7.2144E-01	1.1226	1.9063E+09	1.9067E+03	38.81%)	
6	**sequential**	-2.7243E-01	0.1032	7.4802E+07	74.6327	1.52%)	
7	**combinational**	6.3211	5.4694	1.1391E+09	1.1509E+03	23.42%)	
	Total (uW)	**1.7865E+03**	**6.6952**	**3.1202e+09 pW**	**4.9134E+03**		
		Total Dynamic Power	**3.1202 mW**				

NLDM_COMPILE_ULTRA-1			
S.No	**Design**	**Wire Load Model**	**Library**
1	**picorv32a**	35000	saed32rvt_tt0p78vn4
2	**picorv32_pcpi_fast_mul**	16000	saed32rvt_tt0p78vn4

Cell Internal Power = 1.1339 mW (99%)
Net Switching Power = 6.5549 uW (1%)

Total Dynamic Power = 1.1404 mW (100%)
Cell Leakage Power = 471.1416 uW

NLDM_COMPILE_ULTRA-2							
S.No	**Power Group**	**Internal Power**	**Switching Power**	**Leakage Power**	**Total Power**	**Percentage**	**Attributes**
1	**io_pad**	0.0000	0.0000	0.0000	0.0000	0.00%	
2	**memory**	0.0000	0.0000	0.0000	0.0000	0.00%	
3	**black_box**	0.0000	0.0000	0.0000	0.0000	0.00%	
4	**clock_network**	1.1228E+03	0.0000	0.0000	1.1228E+03	69.67%	I
5	**register**	2.4453	1.0067	3.9237E+08	395.8158	24.56%	
6	**sequential**	0.4972	8.976E-02	1.1333E+06	1.7203	0.11%	
7	**combinational**	8.0849	5.4584	7.7642E+07	91.1855	5.66%	
	Total (uW)	**1.1339E+03**	**6.5549**	**4.7114e+08 pW**	**1.6116E+03**		
		Total Dynamic Power	**1.1404 mW**				

NLDM_COMPILE(MAX)-1

S.No		
1	Startpoint:	genblk1.pcpi_mul/rs2_reg[1] (rising edge-triggered flip-flop clocked
2	Endpoint:	genblk1.pcpi_mul/rd_reg[63] (rising edge-triggered flip-flop clocked by clk)
3	Path Group:	clk
4	Path Type:	Max

NLDM_COMPILE(MAX)-2

Des/Clust/Port	Wire Load Model	Library
picorv32a	35000	saed32rvt_tt0p78vn40c
picorv32_pcpi_fast_mul	8000	saed32rvt_tt0p78vn40c
picorv32_pcpi_fast_mul_DW_mult_tc_0	8000	saed32rvt_tt0p78vn40c

NLDM_COMPILE(MAX)-3

S.No	Point	Incr	Path	A
1	clock clk (rise edge)	0.00	0.00	
2	clock network delay (ideal)	1.00	1.00	
3	genblk1.pcpi_mul/rs2_reg[1]/CLK (DFFX1_LVT)	0.00 #	1.00	r
4	genblk1.pcpi_mul/rs2_reg[1]/Q (DFFX1_LVT)	0.11	1.11	r
5	genblk1.pcpi_mul/mult_2299/b[1] (picorv32_pcpi_fast_mul_DW_mult_tc_0)	0.00	1.11	r
6	genblk1.pcpi_mul/mult_2299/U2255/C1 (HADDX2_LVT)	0.07	1.18	r
7	genblk1.pcpi_mul/mult_2299/U2251/Y (NAND2X0_LVT)	0.03	1.22	f
8	genblk1.pcpi_mul/mult_2299/U2254/Y (NAND3X0_LVT)	0.05	1.27	r
9	genblk1.pcpi_mul/mult_2299/U2073/CO (FADDX1_LVT)	0.10	1.36	r
10	genblk1.pcpi_mul/mult_2299/U2072/CO (FADDX1_LVT)	0.09	1.46	r
11	genblk1.pcpi_mul/mult_2299/U2071/CO (FADDX1_LVT)	0.09	1.55	r
12	genblk1.pcpi_mul/mult_2299/U2070/CO (FADDX1_LVT)	0.09	1.65	r
13	genblk1.pcpi_mul/mult_2299/U2069/CO (FADDX1_LVT)	0.09	1.74	r
14	genblk1.pcpi_mul/mult_2299/U2068/CO (FADDX1_LVT)	0.09	1.84	r
15	genblk1.pcpi_mul/mult_2299/U2067/CO (FADDX1_LVT)	0.09	1.93	r
16	genblk1.pcpi_mul/mult_2299/U2066/CO (FADDX1_LVT)	0.09	2.03	r
17	genblk1.pcpi_mul/mult_2299/U2065/CO (FADDX1_LVT)	0.09	2.12	r
18	genblk1.pcpi_mul/mult_2299/U2064/CO (FADDX1_LVT)	0.09	2.22	r
19	genblk1.pcpi_mul/mult_2299/U2063/S (FADDX1_LVT)	0.15	2.37	f
20	genblk1.pcpi_mul/mult_2299/U2706/Y (AO221X1_LVT)	0.10	2.47	f
21	genblk1.pcpi_mul/mult_2299/U2705/Y (XOR2X1_LVT)	0.10	2.57	r
22	genblk1.pcpi_mul/mult_2299/U315/CO (FADDX1_LVT)	0.11	2.68	r
23	genblk1.pcpi_mul/mult_2299/U314/CO (FADDX1_LVT)	0.09	2.77	r
24	genblk1.pcpi_mul/mult_2299/U313/CO (FADDX1_LVT)	0.09	2.87	r
25	genblk1.pcpi_mul/mult_2299/U312/CO (FADDX1_LVT)	0.09	2.96	r
26	genblk1.pcpi_mul/mult_2299/U311/CO (FADDX1_LVT)	0.09	3.06	r
27	genblk1.pcpi_mul/mult_2299/U310/CO (FADDX1_LVT)	0.09	3.15	r
28	genblk1.pcpi_mul/mult_2299/U309/CO (FADDX1_LVT)	0.09	3.25	r
29	genblk1.pcpi_mul/mult_2299/U308/CO (FADDX1_LVT)	0.09	3.34	r
30	genblk1.pcpi_mul/mult_2299/U307/CO (FADDX1_LVT)	0.09	3.44	r
31	genblk1.pcpi_mul/mult_2299/U306/CO (FADDX1_LVT)	0.09	3.53	r
32	genblk1.pcpi_mul/mult_2299/U305/CO (FADDX1_LVT)	0.09	3.63	r
33	genblk1.pcpi_mul/mult_2299/U304/CO (FADDX1_LVT)	0.09	3.72	r
34	genblk1.pcpi_mul/mult_2299/U303/CO (FADDX1_LVT)	0.09	3.82	r
35	genblk1.pcpi_mul/mult_2299/U302/CO (FADDX1_LVT)	0.09	3.91	r
36	genblk1.pcpi_mul/mult_2299/U301/CO (FADDX1_LVT)	0.09	4.01	r
37	genblk1.pcpi_mul/mult_2299/U300/CO (FADDX1_LVT)	0.09	4.10	r
38	genblk1.pcpi_mul/mult_2299/U299/CO (FADDX1_LVT)	0.09	4.20	r
39	genblk1.pcpi_mul/mult_2299/U298/CO (FADDX1_LVT)	0.09	4.29	r
40	genblk1.pcpi_mul/mult_2299/U297/CO (FADDX1_LVT)	0.09	4.38	r
41	genblk1.pcpi_mul/mult_2299/U296/CO (FADDX1_LVT)	0.09	4.48	r
42	genblk1.pcpi_mul/mult_2299/U295/CO (FADDX1_LVT)	0.09	4.57	r
43	genblk1.pcpi_mul/mult_2299/U294/CO (FADDX1_LVT)	0.09	4.67	r
44	genblk1.pcpi_mul/mult_2299/U293/CO (FADDX1_LVT)	0.09	4.76	r
45	genblk1.pcpi_mul/mult_2299/U292/CO (FADDX1_LVT)	0.09	4.86	r
46	genblk1.pcpi_mul/mult_2299/U291/CO (FADDX1_LVT)	0.09	4.95	r
47	genblk1.pcpi_mul/mult_2299/U290/CO (FADDX1_LVT)	0.09	5.05	r
48	genblk1.pcpi_mul/mult_2299/U289/CO (FADDX1_LVT)	0.09	5.14	r
49	genblk1.pcpi_mul/mult_2299/U2293/Y (NAND2X0_LVT)	0.03	5.18	f
50	genblk1.pcpi_mul/mult_2299/U2231/Y (NAND3X0_LVT)	0.05	5.23	r
51	genblk1.pcpi_mul/mult_2299/U2298/Y (NAND2X0_LVT)	0.03	5.26	f
52	genblk1.pcpi_mul/mult_2299/U2300/Y (NAND3X0_LVT)	0.05	5.31	r
53	genblk1.pcpi_mul/mult_2299/U286/CO (FADDX1_LVT)	0.10	5.41	r
54	genblk1.pcpi_mul/mult_2299/U285/CO (FADDX1_LVT)	0.09	5.50	r
55	genblk1.pcpi_mul/mult_2299/U284/CO (FADDX1_LVT)	0.09	5.60	r
56	genblk1.pcpi_mul/mult_2299/U283/CO (FADDX1_LVT)	0.09	5.69	r
57	genblk1.pcpi_mul/mult_2299/U282/CO (FADDX1_LVT)	0.09	5.79	r
58	genblk1.pcpi_mul/mult_2299/U2246/Y (NAND2X0_LVT)	0.03	5.82	f
59	genblk1.pcpi_mul/mult_2299/U2248/Y (NAND3X0_LVT)	0.05	5.87	r
60	genblk1.pcpi_mul/mult_2299/U280/CO (FADDX1_LVT)	0.10	5.97	r
61	genblk1.pcpi_mul/mult_2299/U279/CO (FADDX1_LVT)	0.09	6.06	r
62	genblk1.pcpi_mul/mult_2299/U278/CO (FADDX1_LVT)	0.09	6.16	r
63	genblk1.pcpi_mul/mult_2299/U277/CO (FADDX1_LVT)	0.09	6.25	r
64	genblk1.pcpi_mul/mult_2299/U276/CO (FADDX1_LVT)	0.09	6.35	r
65	genblk1.pcpi_mul/mult_2299/U275/CO (FADDX1_LVT)	0.09	6.44	r
66	genblk1.pcpi_mul/mult_2299/U274/CO (FADDX1_LVT)	0.09	6.54	r
67	genblk1.pcpi_mul/mult_2299/U273/CO (FADDX1_LVT)	0.09	6.63	r
68	genblk1.pcpi_mul/mult_2299/U272/CO (FADDX1_LVT)	0.09	6.73	r
69	genblk1.pcpi_mul/mult_2299/U271/CO (FADDX1_LVT)	0.09	6.82	r
70	genblk1.pcpi_mul/mult_2299/U270/CO (FADDX1_LVT)	0.09	6.91	r
71	genblk1.pcpi_mul/mult_2299/U269/CO (FADDX1_LVT)	0.09	7.01	r
72	genblk1.pcpi_mul/mult_2299/U268/CO (FADDX1_LVT)	0.09	7.10	r
73	genblk1.pcpi_mul/mult_2299/U267/CO (FADDX1_LVT)	0.09	7.20	r
74	genblk1.pcpi_mul/mult_2299/U266/CO (FADDX1_LVT)	0.09	7.29	r
75	genblk1.pcpi_mul/mult_2299/U2151/Y (XOR3X2_LVT)	0.06	7.35	r
76	genblk1.pcpi_mul/mult_2299/product[63] (picorv32_pcpi_fast_mul_DW_mult_tc_0)	0.00	7.35	r
77	genblk1.pcpi_mul/rd_reg[63]/D (DFFX1_LVT)	0.01	7.36	r
78	data arrival time		7.36	
79	clock clk (rise edge)	6.00	6.00	
80	clock network delay (ideal)	1.00	7.00	
81	clock uncertainty	-1.00	6.00	
82	genblk1.pcpi_mul/rd_reg[63]/CLK (DFFX1_LVT)	0.00	6.00	r
83	library setup time	-0.07	5.93	
84	data required time		5.93	
86	data required time		**5.93**	
87	data arrival time		**-7.36**	
	slack (VIOLATED)		**-1.43**	

NLDM_COMPILE(MIN)-1

S.No		
1	Startpoint:	instr_lw_reg (rising edge-triggered flip-flop clocked by clk)
2	Endpoint:	is_lbu_lhu_lw_reg (rising edge-triggered flip-flop clocked by clk)
3	Path Group:	Clk
4	Path Type:	Min

NLDM_COMPILE(MIN)-2

Des/Clust/Port	Wire Load Model	Library
picorv32a	35000	saed32rvt_tt0p78vn40c

NLDM_COMPILE(MIN)-3

S.No	Point	Incr	Path	
1	clock clk (rise edge)	0.00	0.00	
2	clock network delay (ideal)	0.00	0.00	
3	instr_lw_reg/CLK (DFFX1_LVT)	0.00 #	0.00	r
4	instr_lw_reg/QN (DFFX1_LVT)	0.06	0.06	f
5	U3054/Y (NAND3X0_LVT)	0.04	0.10	r
6	is_lbu_lhu_lw_reg/D (DFFX1_LVT)	0.01	0.11	r
7	data arrival time		0.11	
8	clock clk (rise edge)	0.00	0.00	
9	clock network delay (ideal)	0.00	0.00	
10	is_lbu_lhu_lw_reg/CLK (DFFX1_LVT)	0.00	0.00	r
11	library hold time	-0.02	-0.02	
	data required time		-0.02	
12	data required time		-0.02	
13	data arrival time		-0.11	
	slack (MET)		**0.13**	

NLDM_COMPILE_ULTRA(MAX)-1

S.No		
1	Startpoint:	genblk1.pcpi_mul/rs2_reg[15] (rising edge-triggered flip-flop clocked by clk)
2	Endpoint:	genblk1.pcpi_mul/rd_reg[63] (rising edge-triggered flip-flop clocked by clk)
3	Path Group:	clk
4	Path Type:	max

NLDM_COMPILE_ULTRA(MAX)-2

Des/Clust/Port	Wire Load Model	Library
picorv32a	35000	saed32rvt_tt0p78vn40c
picorv32_pcpi_fast_mul	16000	saed32rvt_tt0p78vn40c

NLDM_COMPILE_ULTRA(MAX)-3

S.No	Point	Incr	Path	
1	clock clk (rise edge)	0.00	0.00	
2	clock network delay (ideal)	1.00	1.00	
3	genblk1.pcpi_mul/rs2_reg[15]/CLK (DFFX1_RVT)	0.00 #	1.00	r
4	genblk1.pcpi_mul/rs2_reg[15]/Q (DFFX1_RVT)	0.30	1.30	f
5	genblk1.pcpi_mul/U346/Y (NBUFFX4_LVT)	0.08	1.38	f
6	genblk1.pcpi_mul/U513/Y (NAND2X0_RVT)	0.10	1.48	r
7	genblk1.pcpi_mul/U262/Y (INVX0_RVT)	0.05	1.53	f
8	genblk1.pcpi_mul/U514/Y (NOR2X0_RVT)	0.11	1.64	r
9	genblk1.pcpi_mul/U515/Y (OA21X1_RVT)	0.08	1.72	r
10	genblk1.pcpi_mul/U249/Y (INVX0_RVT)	0.05	1.77	f
11	genblk1.pcpi_mul/U247/Y (AOI21X1_RVT)	0.18	1.96	r
12	genblk1.pcpi_mul/U539/Y (OA21X1_RVT)	0.12	2.08	r
13	genblk1.pcpi_mul/U79/Y (OA21X1_RVT)	0.11	2.18	r
14	genblk1.pcpi_mul/U75/Y (XNOR2X1_RVT)	0.17	2.36	f
15	genblk1.pcpi_mul/U10/Y (NBUFFX2_RVT)	0.10	2.45	f
16	genblk1.pcpi_mul/U2064/Y (OAI21X1_RVT)	0.18	2.63	r
17	genblk1.pcpi_mul/U2065/Y (XOR2X1_RVT)	0.19	2.82	f
18	genblk1.pcpi_mul/U2110/S (FADDX1_RVT)	0.24	3.06	r
19	genblk1.pcpi_mul/U2103/S (FADDX1_RVT)	0.24	3.30	f
20	genblk1.pcpi_mul/U2075/Y (XOR3X1_RVT)	0.29	3.59	f
21	genblk1.pcpi_mul/U2156/S (FADDX1_RVT)	0.24	3.83	r
22	genblk1.pcpi_mul/U2076/Y (NOR2X0_RVT)	0.12	3.95	f
23	genblk1.pcpi_mul/U2320/Y (OAI21X1_RVT)	0.16	4.11	r
24	genblk1.pcpi_mul/U2323/Y (AOI21X1_RVT)	0.12	4.23	f
25	genblk1.pcpi_mul/U18/Y (OAI21X1_RVT)	0.12	4.35	r
26	genblk1.pcpi_mul/U2338/Y (AO21X1_RVT)	0.10	4.45	r
27	genblk1.pcpi_mul/U2339/Y (AO21X1_RVT)	0.08	4.53	r
28	genblk1.pcpi_mul/U2341/Y (AO22X1_RVT)	0.10	4.64	r
29	genblk1.pcpi_mul/U20/CO (FADDX1_LVT)	0.09	4.73	r
30	genblk1.pcpi_mul/U19/CO (FADDX1_LVT)	0.10	4.83	r
31	genblk1.pcpi_mul/U402/CO (FADDX1_LVT)	0.10	4.93	r
32	genblk1.pcpi_mul/U2343/Y (AO22X1_RVT)	0.11	5.04	r
33	genblk1.pcpi_mul/U403/CO (FADDX1_LVT)	0.12	5.16	r
34	genblk1.pcpi_mul/U416/Y (AO22X1_LVT)	0.07	5.23	r
35	genblk1.pcpi_mul/U404/CO (FADDX1_LVT)	0.11	5.34	r
36	genblk1.pcpi_mul/U2347/Y (AO22X1_RVT)	0.11	5.45	r
37	genblk1.pcpi_mul/U2357/CO (FADDX1_RVT)	0.18	5.63	r
38	genblk1.pcpi_mul/U2369/Y (AO22X1_RVT)	0.11	5.75	r
39	genblk1.pcpi_mul/U2371/Y (AO22X1_RVT)	0.10	5.85	r
40	genblk1.pcpi_mul/U2377/Y (XOR2X1_LVT)	0.06	5.91	r
41	genblk1.pcpi_mul/rd_reg[63]/D (DFFX1_RVT)	0.01	5.92	r
42	data arrival time		5.92	
43	clock clk (rise edge)	6.00	6.00	
44	clock network delay (ideal)	1.00	7.00	
45	clock uncertainty	-1.00	6.00	
46	genblk1.pcpi_mul/rd_reg[63]/CLK (DFFX1_RVT)	0.00	6.00	r
47	library setup time	-0.08	5.92	
48	data required time		5.92	
49	data required time		5.92	
50	data arrival time		-5.92	
	Slack (MET)		**0.00**	

NLDM_COMPILE_ULTRA(MIN)-1

S.No		
1	Startpoint:	count_cycle_reg[0] (rising edge-triggered flip-flop clocked by clk)
2	Endpoint:	count_cycle_reg[0] (rising edge-triggered flip-flop clocked by clk)
3	Path Group:	Clk
4	Path Type:	Min

NLDM_COMPILE_ULTRA(MIN)-2

Des/Clust/Port	Wire Load Model	Library
picorv32a	35000	saed32rvt_tt0p78vn40c

NLDM_COMPILE_ULTRA(MIN)-3

S.No	Point	Incr	Path	
1	clock clk (rise edge)	0.00	0.00	
2	clock network delay (ideal)	0.00	0.00	
3	count_cycle_reg[0]/CLK (DFFX1_LVT)	0.00 #	0.00	r
4	count_cycle_reg[0]/QN (DFFX1_LVT)	0.06	0.06	f
5	U9880/Y (AND2X1_RVT)	0.07	0.13	f
6	count_cycle_reg[0]/D (DFFX1_LVT)	0.01	0.14	f
7	data arrival time		0.14	
8	clock clk (rise edge)	0.00	0.00	
9	clock network delay (ideal)	0.00	0.00	
10	is_lbu_lhu_lw_reg/CLK (DFFX1_LVT)	0.00	0.00	r
11	library hold time	-0.02	-0.02	
	data required time		-0.02	
12	data required time		-0.02	
13	data arrival time		-0.14	
	slack (MET)		**0.16**	

NLDM_COMPILE		
S.No	**Timing Path Group (none)**	
1	Levels of Logic:	70.00
2	Critical Path Length:	6.36
3	Critical Path Slack:	-1.43
4	Critical Path Clk Period:	6.00
5	Total Negative Slack:	-13.62
6	No. of Violating	19.00
7	Worst Hold	0.00
8	Total Hold Violation:	0.00
9	No. of Hold	0.00
	Cell Count	
10	Hierarchical Cell	15
11	Hierarchical Port	1510
12	Leaf Cell Count:	8473
13	Buf/Inv Cell Count:	1143
14	Buf Cell Count:	26
15	Inv Cell Count:	1117
16	CT Buf/Inv Cell	0
17	Combinational Cell Count:	6806
18	Sequential Cell Count:	1667
19	Macro Count:	0
	Area	
20	Combinational Area:	18679.329838
21	Noncombinational Area:	10927.684057
22	Buf/Inv Area:	1580.521542
23	Total Buffer Area:	98.61
24	Total Inverter Area:	1481.91
25	Macro/Black Box	0.000000
26	Net Area:	8977.222396
	Cell Area:	**29607.013896**
	Design Area:	**38584.236291**
	Design Rules	
	Total Number of Nets:	10245
	Nets With Violations:	0
	Max Trans	0
	Max Cap Violations:	0

NLDM_COMPILE_ULTRA		
S.No	**Timing Path Group (none)**	
1	Levels of Logic:	36.00
2	Critical Path Length:	4.92
3	Critical Path Slack:	0.00
4	Critical Path Clk Period:	6.00
5	Total Negative Slack:	0.00
6	No. of Violating	0.00
7	Worst Hold	0.00
8	Total Hold Violation:	0.00
9	No. of Hold	0.00
	Cell Count	
10	Hierarchical Cell	1
11	Hierarchical Port	134
12	Leaf Cell Count:	10835
13	Buf/Inv Cell Count:	1224
14	Buf Cell Count:	59
15	Inv Cell Count:	1165
16	CT Buf/Inv Cell	0
17	Combinational Cell Count:	9185
18	Sequential Cell Count:	1650
19	Macro Count:	0
	Area	
20	Combinational Area:	23998.309593
21	Noncombinational Area:	10804.932503
22	Buf/Inv Area:	1639.482953
23	Total Buffer Area:	127.58
24	Total Inverter Area:	1511.90
25	Macro/Black Box	0.000000
26	Net Area:	11317.279926
	Cell Area:	**34803.242096**
	Design Area:	**46120.522022**
	Design Rules	
	Total Number of Nets:	11858
	Nets With Violations:	0
	Max Trans	0
	Max Cap Violations:	0

NLDM_COMPILE 35000

S.No	Fanout	Lengths	Points	Average cap	Standard Deviation	%SD
1	1	16.92				
2	2	39.03				
3	3	64.09				
4	4	92.51				
5	5	124.73				
6	6	161.17				
7	7	202.25				
8	8	248.42				
9	9	300.09				
10	10	357.68				
11	11	421.63				
12	12	492.37				
13	13	570.31				
14	14	655.89				
15	15	749.54				
16	16	851.67				
17	17	962.72				
18	18	1083.11				
19	19	1213.28				
20	20	1353.64				
	Total Length					

NLDM_COMPILE 8000

S.No	Fanout	Lengths	Points	Average cap	Standard Deviation	%SD
1	1	13.94				
2	2	31.80				
3	3	51.61				
4	4	73.61				
5	5	98.05				
6	6	125.17				
7	7	155.23				
8	8	188.46				
9	9	225.12				
10	10	265.45				
11	11	309.71				
12	12	358.13				
13	13	410.96				
14	14	468.46				
15	15	530.86				
16	16	598.42				
17	17	671.38				
18	18	749.98				
19	19	834.49				
20	20	925.13				
	Total Length					

NLDM_COMPILE- FORQA

S.No	Fanout	Lengths	Points	Average cap	Standard Deviation	%SD
1	1	8.28				
2	2	18.49				
3	3	29.35				
4	4	40.92				
5	5	53.23				
6	6	66.36				
7	7	80.36				
8	8	95.27				
9	9	111.17				
10	10	128.09				
11	11	146.10				
12	12	165.26				
13	13	185.61				
14	14	207.22				
15	15	230.13				
16	16	254.41				
17	17	280.11				
18	18	307.28				
19	19	335.98				
20	20	366.27				
	Total Length					

NLDM_COMPILE_ULTRA -1

Wire load model:	35000
Location :	Picorv32a (design)
Resistance :	0.001187
Capacitance :	0.00027
Area :	0.01
Slope :	140.36

NLDM_COMPILE_ULTRA-2

Wire load model:	16000
Location :	picorv32_pcpi_fast_mul (design)
Resistance :	0.00128
Capacitance :	0.00056
Area :	0.01
Slope :	108.438

NLDM_COMPILE_ULTRA -35000

S.No	Fanout	Lengths	Points	Average cap	Standard Deviation	%SD
1	1	16.92				
2	2	39.03				
3	3	64.09				
4	4	92.51				
5	5	124.73				
6	6	161.17				
7	7	202.25				
8	8	248.42				
9	9	300.09				
10	10	357.68				
11	11	421.63				
12	12	492.37				
13	13	570.31				
14	14	655.89				
15	15	749.54				
16	16	851.67				
17	17	962.72				
18	18	1083.11				
19	19	1213.28				
20	20	1353.64				
	Total Length					

NLDM_COMPILE_ULTRA - 16000

S.No	Fanout	Lengths	Points	Average cap	Standard Deviation	%SD
1	1	15.10				
2	2	34.61				
3	3	56.42				
4	4	80.84				
5	5	108.20				
6	6	138.79				
7	7	172.92				
8	8	210.91				
9	9	253.07				
10	10	299.71				
11	11	351.13				
12	12	407.64				
13	13	469.57				
14	14	537.20				
15	15	610.87				
16	16	690.86				
17	17	777.51				
18	18	871.11				
19	19	971.97				
20	20	1080.41				
	Total Length					

NLDM_COMPILE -1

Wire load model	35000
Location	Picorv32a (design)
Resistance	0.001187
Capacitance	0.00027
Area	0.01
Slope	140.36

NLDM_COMPILE- 2

Wire load model:	8000
Location	picorv32_pcpi_fast_mul (design)
Resistance :	0.0015727
Capacitance :	0.000312
Area :	0.01
Slope :	90.6464

NLDM_COMPILE -3

Wire load model:	ForQA
Location :	picorv32a_DW01_add_0 (design)
Resistance :	0.002067
Capacitance :	0.026724
Area :	0.01
Slope :	30.2854

NLDM_COMPILE				
S.No	**Vth Group Name->**	**saed32cell_lvt**	**saed32cell_svt**	**Total**
2	**All Cells**	7619 (89.92%)	854 (10.08%)	8473 (100.00%)
3	**Blackbox Cells**	0 (0.00%)	0 (0.00%)	0 (0.00%)
4	**Non-Blackbox cells**	7619 (89.92%)	854 (10.08%)	8473 (100.00%)
5	**All Cells Area**	26258.41 (88.69%)	3348.60 (11.31%)	29607.01 (100.00%)
6	**Blackbox Cells area**	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)
7	**Non-Blackbox cells area**	26258.41 (88.69%)	3348.60 (11.31%)	29607.01 (100.00%)
8	**All Cells leakage**	3.114mW (99.80%)	6.150uW (0.20%)	3.120mW (100.00%)
9	**Blackbox Cells leakage**	0.000pW (0.00%)	0.000pW (0.00%)	0.000pW (0.00%)
10	**Non-Blackbox cells leakage**	3.114mW (99.80%)	6.150uW (0.20%)	3.120mW (100.00%)

NLDM_COMPILE_ULTRA				
S.No	**Vth Group Name->**	**saed32cell_lvt**	**saed32cell_svt**	**Total**
2	**All Cells**	499 (4.61%)	10336 (95.39%)	10835
3	**Blackbox Cells**	0 (0.00%)	0 (0.00%)	0
4	**Non-Blackbox cells**	499 (4.61%)	10336 (95.39%)	10835
5	**All Cells Area**	2375.99 (6.83%)	32427.25 (93.17%)	34803.24 (100.00%)
6	**Blackbox Cells area**	0 (0.00%)	0 (0.00%)	0
7	**Non-Blackbox cells area**	2375.99 (6.83%)	32427.25 (93.17%)	34803.24 (100.00%)
8	**All Cells leakage**	414.344uW (87.94%)	56.797uW (12.06%)	471.142uW (100.00%)
9	**Blackbox Cells leakage**	0 (0.00%)	0 (0.00%)	0
10	**Non-Blackbox cells leakage**	414.344uW (87.94%)	56.797uW (12.06%)	471.142uW (100.00%)

NLDM_COMPILE Global Skew Report

S.No		
1	Clock Tree Name :	"clk"
2	Clock Period :	6
3	Clock Tree root pin :	"clk"
4	Number of Levels :	1
5	Number of Sinks :	1599
6	Number of CT Buffers :	0
7	Number of CTS added gates :	0
8	Number of Preexisting Gates :	0
9	Number of Preexisting Buf/Inv :	0
10	Total Number of Clock Cells :	0
11	Total Area of CT Buffers :	0
12	Total Area of CT cells :	0
13	Max Global Skew :	0.00173
14	Number of MaxTran Violators :	0
15	Number of MaxCap Violators :	1
16	Number of MaxFanout Violators :	0
1	Operating Condition	worst
2	Clock global Skew	0.002
3	Longest path delay	0.125
4	Shortest path delay	0.123

The longest path delay end pin: genblk1.pcpi_mul/rd_reg[39]/CLK
The shortest path delay end pin: genblk1.pcpi_mul/shift_out_reg/CLK

Clock_tree Longest Path (Compile ultra)

S.No	Pin	Cap	Fanout	Trans	Incr	Arri	
1	clk	0	1	0	0	0	r
2	clk	1001037.500	1599	0	0	0	r
3	genblk1.pcpi_mul/rd_reg[39]/CLK	1001037.500	0	0	0.125	0.125	r
	Clock Delay					0.125	

Clock_tree Shortest Path (Compile ultra)

S.No	Pin	Cap	Fanout	Trans	Incr	Arri	
1	clk	0	1	0	0	0	r
2	clk	1001037.500	1599	0	0	0	r
3	genblk1.pcpi_mul/shift_out_reg/CLK	1001037.500	0	0	0.123	0.123	r
	Clock Delay					0.123	

NLDM_COMPILE_ULTRA Global Skew

S.No		
1	Clock Tree Name :	"clk"
2	Clock Period :	6
3	Clock Tree root pin :	"clk"
4	Number of Levels :	1
5	Number of Sinks :	1582
6	Number of CT Buffers :	0
7	Number of CTS added gates :	0
8	Number of Preexisting Gates :	0
9	Number of Preexisting Buf/Inv :	0
10	Total Number of Clock Cells :	0
11	Total Area of CT Buffers :	0
12	Total Area of CT cells :	0
13	Max Global Skew :	0.00247
14	Number of MaxTran Violators :	0
15	Number of MaxCap Violators :	1
16	Number of MaxFanout Violators :	0
1	Operating Condition	worst
2	Clock global Skew	0.002
3	Longest path delay	0.125
4	Shortest path delay	0.123

The longest path delay end pin: genblk1.pcpi_mul/rs2_reg[3]/CLK
The shortest path delay end pin: genblk1.pcpi_mul/rs1_reg[32]/CLK

NLDM_COMPILE_ULTRA Clock_tree Longest Path

S.No	Pin	Cap	Fanout	Trans	Incr	Arri	
1	clk	0	1	0	0	0	r
2	clk	1001037.500	1582	0	0	0	r
3	genblk1.pcpi_mul/rs2_reg[3]/CLK	1001037.500	0	0	0.125	0.125	r
	Clock Delay					0.125	

NLDM_COMPILE_ULTRA Clock_tree Shortest Path

S.No	Pin	Cap	Fanout	Trans	Incr	Arri	
1	clk	0	1	0	0	0	r
2	clk	1001037.500	1582	0	0	0	r
3	genblk1.pcpi_mul/rs1_reg[32]/CLK	1001037.500	0	0	0.123	0.123	r
	Clock Delay					0.123	

hierarchy

Attributes:	
r	: licensed design

NLDM_COMPILE

picorv32a	r
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
AND2X1_RVT	sscd32rvt_ntfp78rvt40c
AND2X2_LVT	sscd32rvt_ntfp78rvt40c
AND3X1_LVT	sscd32rvt_ntfp78rvt40c
AND3X2_LVT	sscd32rvt_ntfp78rvt40c
AND4X1_LVT	sscd32rvt_ntfp78rvt40c
AND4X2_LVT	sscd32rvt_ntfp78rvt40c
AO21X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_RVT	sscd32rvt_ntfp78rvt40c
AO221X1_LVT	sscd32rvt_ntfp78rvt40c
AO221X1_RVT	sscd32rvt_ntfp78rvt40c
AO221X2_LVT	sscd32rvt_ntfp78rvt40c
AO222X1_LVT	sscd32rvt_ntfp78rvt40c
AO31X1_LVT	sscd32rvt_ntfp78rvt40c
AO322X1_LVT	sscd32rvt_ntfp78rvt40c
AO322X1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_RVT	sscd32rvt_ntfp78rvt40c
DFFX2_LVT	sscd32rvt_ntfp78rvt40c
INVX0_LVT	sscd32rvt_ntfp78rvt40c
INVX0_RVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
INVX2_LVT	sscd32rvt_ntfp78rvt40c
INVX4_LVT	sscd32rvt_ntfp78rvt40c
LATCHX1_LVT	sscd32rvt_ntfp78rvt40c
LATCHX1_RVT	sscd32rvt_ntfp78rvt40c
MUX21X1_LVT	sscd32rvt_ntfp78rvt40c
MUX21X1_RVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_RVT	sscd32rvt_ntfp78rvt40c
NAND2X2_LVT	sscd32rvt_ntfp78rvt40c
NAND3X0_LVT	sscd32rvt_ntfp78rvt40c
NAND3X0_RVT	sscd32rvt_ntfp78rvt40c
NAND4X0_LVT	sscd32rvt_ntfp78rvt40c
NAND4X0_RVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_LVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_RVT	sscd32rvt_ntfp78rvt40c
NOR2X0_LVT	sscd32rvt_ntfp78rvt40c
NOR3X0_LVT	sscd32rvt_ntfp78rvt40c
NOR4X1_LVT	sscd32rvt_ntfp78rvt40c
OA21X1_LVT	sscd32rvt_ntfp78rvt40c
OA22X1_LVT	sscd32rvt_ntfp78rvt40c
OA221X1_LVT	sscd32rvt_ntfp78rvt40c
OA222X1_LVT	sscd32rvt_ntfp78rvt40c
OA21X1_RVT	sscd32rvt_ntfp78rvt40c
OA22X1_RVT	sscd32rvt_ntfp78rvt40c
OA221X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c
OR3X1_LVT	sscd32rvt_ntfp78rvt40c
OR4X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32_pcpi_fast_mul	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
AND4X1_LVT	sscd32rvt_ntfp78rvt40c
AO21X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_RVT	sscd32rvt_ntfp78rvt40c
DFFX1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_RVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
INVX2_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NAND3X0_LVT	sscd32rvt_ntfp78rvt40c
NAND4X0_LVT	sscd32rvt_ntfp78rvt40c
NOR3X0_LVT	sscd32rvt_ntfp78rvt40c
NOR4X1_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c
OR3X1_LVT	sscd32rvt_ntfp78rvt40c
picorv32_pcpi_fast_mul_DW_mult_tc_0	r
AND2X4_RVT	sscd32rvt_ntfp78rvt40c
AND3X4_LVT	sscd32rvt_ntfp78rvt40c
AO21X1_LVT	sscd32rvt_ntfp78rvt40c
AO21X1_RVT	sscd32rvt_ntfp78rvt40c
AO21X2_RVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_RVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_RVT	sscd32rvt_ntfp78rvt40c
AO22X2_RVT	sscd32rvt_ntfp78rvt40c
AO221X1_LVT	sscd32rvt_ntfp78rvt40c
AO221X1_RVT	sscd32rvt_ntfp78rvt40c
AO222X1_LVT	sscd32rvt_ntfp78rvt40c
AO222X1_RVT	sscd32rvt_ntfp78rvt40c
DELLN1X2_LVT	sscd32rvt_ntfp78rvt40c
DELLN2X2_RVT	sscd32rvt_ntfp78rvt40c
DELLN3X2_RVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
HADD1X1_LVT	sscd32rvt_ntfp78rvt40c
HADD1X2_LVT	sscd32rvt_ntfp78rvt40c
IBUFFX2_LVT	sscd32rvt_ntfp78rvt40c
IBUFFX2_RVT	sscd32rvt_ntfp78rvt40c
IBUFFX8_LVT	sscd32rvt_ntfp78rvt40c
IBUFFX16_LVT	sscd32rvt_ntfp78rvt40c
INVX0_LVT	sscd32rvt_ntfp78rvt40c
INVX0_RVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
INVX2_LVT	sscd32rvt_ntfp78rvt40c
INVX2_RVT	sscd32rvt_ntfp78rvt40c
INVX4_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_RVT	sscd32rvt_ntfp78rvt40c
NAND3X0_LVT	sscd32rvt_ntfp78rvt40c
NAND3X0_RVT	sscd32rvt_ntfp78rvt40c
NAND4X0_RVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_LVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_RVT	sscd32rvt_ntfp78rvt40c
NOR2X0_LVT	sscd32rvt_ntfp78rvt40c
NOR2X0_RVT	sscd32rvt_ntfp78rvt40c
OA21X1_LVT	sscd32rvt_ntfp78rvt40c
OA21X1_RVT	sscd32rvt_ntfp78rvt40c
OA22X1_LVT	sscd32rvt_ntfp78rvt40c
OA22X1_RVT	sscd32rvt_ntfp78rvt40c
OA221X1_LVT	sscd32rvt_ntfp78rvt40c
OA221X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X2_LVT	sscd32rvt_ntfp78rvt40c
OR2X2_RVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_RVT	sscd32rvt_ntfp78rvt40c
XNOR2X2_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
XOR2X2_LVT	sscd32rvt_ntfp78rvt40c
XOR3X1_LVT	sscd32rvt_ntfp78rvt40c
XOR3X2_LVT	sscd32rvt_ntfp78rvt40c
XOR3X2_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_0	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_1	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_2	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_3	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_4	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_5	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_add_6	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_cmp6_0	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
AO221X1_LVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_RVT	sscd32rvt_ntfp78rvt40c
NAND4X0_LVT	sscd32rvt_ntfp78rvt40c
NOR4X0_LVT	sscd32rvt_ntfp78rvt40c
OA22X1_LVT	sscd32rvt_ntfp78rvt40c
OA221X1_RVT	sscd32rvt_ntfp78rvt40c
OA222X1_LVT	sscd32rvt_ntfp78rvt40c
OA222X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_dec_0_DW01_dec_1	
AO21X1_LVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NOR2X0_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_inc_0	
HADD1X1_LVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_inc_1	
HADD1X1_LVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW01_sub_0	
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
INVX0_RVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_RVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_RVT	sscd32rvt_ntfp78rvt40c
picorv32a_DW_cmp_0	r
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
AND4X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO221X1_LVT	sscd32rvt_ntfp78rvt40c
AO222X1_LVT	sscd32rvt_ntfp78rvt40c
INVX0_RVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NAND2X0_RVT	sscd32rvt_ntfp78rvt40c
NAND3X0_LVT	sscd32rvt_ntfp78rvt40c
NAND4X0_LVT	sscd32rvt_ntfp78rvt40c
NOR3X0_LVT	sscd32rvt_ntfp78rvt40c
OA21X1_LVT	sscd32rvt_ntfp78rvt40c
OA22X1_LVT	sscd32rvt_ntfp78rvt40c
OA22X1_RVT	sscd32rvt_ntfp78rvt40c
OA221X1_LVT	sscd32rvt_ntfp78rvt40c
OA221X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c

NLDM_COMPILE_ULTRA

picorv32a	
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
AND2X1_RVT	sscd32rvt_ntfp78rvt40c
AND2X2_LVT	sscd32rvt_ntfp78rvt40c
AND2X2_RVT	sscd32rvt_ntfp78rvt40c
AND2X4_LVT	sscd32rvt_ntfp78rvt40c
AND2X4_RVT	sscd32rvt_ntfp78rvt40c
AND3X1_RVT	sscd32rvt_ntfp78rvt40c
AND3X2_RVT	sscd32rvt_ntfp78rvt40c
AND4X1_LVT	sscd32rvt_ntfp78rvt40c
AND4X1_RVT	sscd32rvt_ntfp78rvt40c
AO21X1_LVT	sscd32rvt_ntfp78rvt40c
AO21X1_RVT	sscd32rvt_ntfp78rvt40c
AO21X2_RVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_RVT	sscd32rvt_ntfp78rvt40c
AO221X1_RVT	sscd32rvt_ntfp78rvt40c
AO221X2_RVT	sscd32rvt_ntfp78rvt40c
AO222X1_LVT	sscd32rvt_ntfp78rvt40c
AO222X1_RVT	sscd32rvt_ntfp78rvt40c
AO31X1_LVT	sscd32rvt_ntfp78rvt40c
AO31X1_RVT	sscd32rvt_ntfp78rvt40c
AO322X1_LVT	sscd32rvt_ntfp78rvt40c
AO322X1_RVT	sscd32rvt_ntfp78rvt40c
DFFSSRX1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_RVT	sscd32rvt_ntfp78rvt40c
DFFX2_LVT	sscd32rvt_ntfp78rvt40c
DFFX2_RVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_RVT	sscd32rvt_ntfp78rvt40c
HADD1X1_RVT	sscd32rvt_ntfp78rvt40c
IBUFFX16_RVT	sscd32rvt_ntfp78rvt40c
INVX0_LVT	sscd32rvt_ntfp78rvt40c
INVX0_RVT	sscd32rvt_ntfp78rvt40c
INVX1_LVT	sscd32rvt_ntfp78rvt40c
INVX1_RVT	sscd32rvt_ntfp78rvt40c
INVX2_LVT	sscd32rvt_ntfp78rvt40c
INVX2_RVT	sscd32rvt_ntfp78rvt40c
INVX4_LVT	sscd32rvt_ntfp78rvt40c
INVX4_RVT	sscd32rvt_ntfp78rvt40c
INVX8_RVT	sscd32rvt_ntfp78rvt40c
LATCHX1_RVT	sscd32rvt_ntfp78rvt40c
MUX21X1_RVT	sscd32rvt_ntfp78rvt40c
NAND2X0_LVT	sscd32rvt_ntfp78rvt40c
NAND2X4_LVT	sscd32rvt_ntfp78rvt40c
NAND3X0_LVT	sscd32rvt_ntfp78rvt40c
NAND3X0_RVT	sscd32rvt_ntfp78rvt40c
NAND3X1_RVT	sscd32rvt_ntfp78rvt40c
NAND3X2_RVT	sscd32rvt_ntfp78rvt40c
NAND4X0_LVT	sscd32rvt_ntfp78rvt40c
NAND4X0_RVT	sscd32rvt_ntfp78rvt40c
NAND4X1_LVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_LVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_RVT	sscd32rvt_ntfp78rvt40c
NBUFFX4_RVT	sscd32rvt_ntfp78rvt40c
NOR2X0_LVT	sscd32rvt_ntfp78rvt40c
NOR2X0_RVT	sscd32rvt_ntfp78rvt40c
NOR2X2_RVT	sscd32rvt_ntfp78rvt40c
NOR2X4_RVT	sscd32rvt_ntfp78rvt40c
NOR3X0_LVT	sscd32rvt_ntfp78rvt40c
NOR3X0_RVT	sscd32rvt_ntfp78rvt40c
NOR3X2_RVT	sscd32rvt_ntfp78rvt40c
NOR4X0_RVT	sscd32rvt_ntfp78rvt40c
NOR4X1_LVT	sscd32rvt_ntfp78rvt40c
OA21X1_LVT	sscd32rvt_ntfp78rvt40c
OA21X1_RVT	sscd32rvt_ntfp78rvt40c
OA22X1_LVT	sscd32rvt_ntfp78rvt40c
OA22X1_RVT	sscd32rvt_ntfp78rvt40c
OA22X2_RVT	sscd32rvt_ntfp78rvt40c
OA221X1_RVT	sscd32rvt_ntfp78rvt40c
OA222X1_RVT	sscd32rvt_ntfp78rvt40c
OA31X1_LVT	sscd32rvt_ntfp78rvt40c
OA31X1_RVT	sscd32rvt_ntfp78rvt40c
OA332X1_LVT	sscd32rvt_ntfp78rvt40c
OA332X1_RVT	sscd32rvt_ntfp78rvt40c
OA333X1_RVT	sscd32rvt_ntfp78rvt40c
OA22X1_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_LVT	sscd32rvt_ntfp78rvt40c
OR2X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X2_RVT	sscd32rvt_ntfp78rvt40c
OR2X4_RVT	sscd32rvt_ntfp78rvt40c
OR3X1_LVT	sscd32rvt_ntfp78rvt40c
OR3X1_RVT	sscd32rvt_ntfp78rvt40c
OR3X2_LVT	sscd32rvt_ntfp78rvt40c
OR3X2_RVT	sscd32rvt_ntfp78rvt40c
OR3X4_RVT	sscd32rvt_ntfp78rvt40c
OR4X1_LVT	sscd32rvt_ntfp78rvt40c
OR4X1_RVT	sscd32rvt_ntfp78rvt40c
OR4X2_RVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_RVT	sscd32rvt_ntfp78rvt40c
XNOR3X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR3X1_RVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
XOR3X1_LVT	sscd32rvt_ntfp78rvt40c
XOR3X1_RVT	sscd32rvt_ntfp78rvt40c
XOR3X2_LVT	sscd32rvt_ntfp78rvt40c
picorv32_pcpi_fast_mul	r
AND2X1_LVT	sscd32rvt_ntfp78rvt40c
AND2X2_RVT	sscd32rvt_ntfp78rvt40c
AND3X1_RVT	sscd32rvt_ntfp78rvt40c
AND3X2_LVT	sscd32rvt_ntfp78rvt40c
AND3X2_RVT	sscd32rvt_ntfp78rvt40c
AO21X1_RVT	sscd32rvt_ntfp78rvt40c
AO22X1_LVT	sscd32rvt_ntfp78rvt40c
AO22X1_RVT	sscd32rvt_ntfp78rvt40c
AO221X1_RVT	sscd32rvt_ntfp78rvt40c
AO222X1_RVT	sscd32rvt_ntfp78rvt40c
DFFSSRX1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_LVT	sscd32rvt_ntfp78rvt40c
DFFX1_RVT	sscd32rvt_ntfp78rvt40c
DFFX2_LVT	sscd32rvt_ntfp78rvt40c
DFFX2_RVT	sscd32rvt_ntfp78rvt40c
FADD1X1_LVT	sscd32rvt_ntfp78rvt40c
FADD1X1_RVT	sscd32rvt_ntfp78rvt40c
HADD1X1_RVT	sscd32rvt_ntfp78rvt40c
INVX0_LVT	sscd32rvt_ntfp78rvt40c
INVX0_RVT	sscd32rvt_ntfp78rvt40c
INVX2_RVT	sscd32rvt_ntfp78rvt40c
INVX8_RVT	sscd32rvt_ntfp78rvt40c
NAND2X0_RVT	sscd32rvt_ntfp78rvt40c
NAND3X0_RVT	sscd32rvt_ntfp78rvt40c
NAND4X0_RVT	sscd32rvt_ntfp78rvt40c
NBUFFX2_RVT	sscd32rvt_ntfp78rvt40c
NBUFFX4_LVT	sscd32rvt_ntfp78rvt40c
NBUFFX4_RVT	sscd32rvt_ntfp78rvt40c
NBUFFX8_RVT	sscd32rvt_ntfp78rvt40c
NOR2X0_RVT	sscd32rvt_ntfp78rvt40c
NOR2X2_RVT	sscd32rvt_ntfp78rvt40c
OA21X1_RVT	sscd32rvt_ntfp78rvt40c
OA22X1_RVT	sscd32rvt_ntfp78rvt40c
OA221X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X1_RVT	sscd32rvt_ntfp78rvt40c
OR2X2_RVT	sscd32rvt_ntfp78rvt40c
OR2X4_RVT	sscd32rvt_ntfp78rvt40c
XNOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR3X1_LVT	sscd32rvt_ntfp78rvt40c
XNOR3X1_RVT	sscd32rvt_ntfp78rvt40c
XOR2X1_LVT	sscd32rvt_ntfp78rvt40c
XOR2X1_RVT	sscd32rvt_ntfp78rvt40c
XOR3X1_LVT	sscd32rvt_ntfp78rvt40c
XOR3X1_RVT	sscd32rvt_ntfp78rvt40c
XOR3X2_LVT	sscd32rvt_ntfp78rvt40c
XOR3X2_RVT	sscd32rvt_ntfp78rvt40c

COMPILE			
Report	**Parameter**	**NLDM**	**CCS**
report_cell	Top Module cells	**5343**	**5458**
get_cells -hierarchical	Total Cells	**8598**	**9327**
report_reference	No of Refernces	**71**	**63**
report_qor	Levels of Logic	**70.00**	**60.00**
	Critical Path Length	**6.36**	**4.93**
	Critical Path Slack	**-1.43**	**0.00**
	Total Negative Slack	**-13.62**	**0.00**
	No. of Violating Paths	**19.00**	**0.00**
report_area	Combinational Area	**18679.329838**	**20642.592267**
	Sequential Area	**1580.521542**	**2197.074930**
	Buf/Inv Area	**10927.684057**	**10911.418844**
	Net Interconnect Area	**8977.222396**	**8570.823434**
	Total area (µm²)	**38584.236291**	**40124.834545**
report_power	Internal Power (µW)	**1.7865E+03**	**1.3476E+03**
	Switching Power (µW)	**6.6952**	**5.8677**
	Total Dynamic Power (mW)	**3.1202**	**1.3535**
	Leakage Power (pW)	**3.1202E+09**	**8.3429E+08**
	Total Power (µW)	**4.9134E+03**	**2.1878E+03**
	SETUP		
report_timing	Data Required Time	**5.93**	**5.93**
	Data Arrival Time	**-7.36**	**-5.93**
	Slack(max) - Setup	**-1.43**	**0.00**
	HOLD		
	Data Required Time	**-0.02**	**-0.02**
	Data Arrival Time	**-0.11**	**-0.19**
	Slack(min) - Hold	**0.13**	**0.21**
report_threshold_voltage_group	**LVT**		
	Cell Count	**7619 (89.92%)**	**7768 (84.23%)**
	Area	**26258.41 (88.69%)**	**23872.76 (75.66%)**
	Leakage (µW)	**3.114 (99.80%)**	**660.927 (79.22%)**
	RVT		
	Cell Count	**854 (10.08%)**	**1454 (15.77%)**
	Area	**3348.60 (11.31%)**	**7681.25 (24.34%)**
	Leakage (µW)	**6.150 (0.20%)**	**173.367 (20.78%)**
report_clock_tree	No of Sinks	**1599**	**1599**
	Clock Global Skew	**0.002**	**0.004**
	Longest Path Delay	**0.125**	**0.113**
	Shortest Path Delay	**0.123**	**0.109**

In compile, ccs is better than nldm in terms of timing and power but a slightly higher in area
NLDM chose more LVT than CCS

COMPILE_ULTRA			
Report	**Parameter**	**NLDM**	**CCS**
report_cell	Top Module cells	**7930**	**6387**
get_cells -hierarchical	Total Cells	**10841**	**9083**
report_reference	No of Refernces	**98**	**39**
report_qor	Levels of Logic	**36.00**	**43.00**
	Critical Path Length	**4.92**	**4.93**
	Critical Path Slack	**0.00**	**0.00**
	Total Negative Slack	**0.00**	**0.00**
	No. of Violating Paths	**0.00**	**0.00**
report_area	Combinational Area	**23998.309593**	**19846.359044**
	Sequential Area	**1639.482953**	**862.564742**
	Buf/Inv Area	**10804.932503**	**10799.595480**
	Net Interconnect Area	**11317.279926**	**12944.777122**
	Total area (µm²)	**46120.522022**	**43590.731646**
report_power	Internal Power (µW)	**1.1339E+03**	**1.8953E+03**
	Switching Power (µW)	**6.5549**	**5.6572**
	Total Dynamic Power (mW)	**1.1404**	**1.9009**
	Leakage Power (pW)	**4.7114E+08**	**8.1519E+08**
	Total Power (µW)	**1.6116E+03**	**2.7161E+03**
	SETUP		
report_timing	Data Required Time	**5.92**	**5.93**
	Data Arrival Time	**-5.92**	**-5.93**
	Slack(max) - Setup	**0.00**	**0.00**
	HOLD		
	Data Required Time	**-0.02**	**-0.02**
	Data Arrival Time	**-0.14**	**-0.16**
	Slack(min) - Hold	**0.16**	**0.18**
report_threshold_voltage_group	**LVT**		
	Cell Count	**499 (4.61%)**	**9056 (99.72%)**
	Area	**2375.99 (6.83%)**	**30607.83 (99.88%)**
	Leakage (µW)	**414.344 (87.94%)**	**813.229(99.76%)**
	RVT		
	Cell Count	**10336 (95.39%)**	**25 (0.28%)**
	Area	**32427.25 (93.17%)**	**38.12 (0.12%)**
	Leakage (µW)	**56.797 (12.06%)**	**1.965 (0.24%)**
report_clock_tree	No of Sinks	**1582**	**1582**
	Clock Global Skew	**0.002**	**0.000**
	Longest Path Delay	**0.125**	**0.113**
	Shortest Path Delay	**0.123**	**0.113**

With Compile_ultra, the ccs is better in terms of timing and area,
But power consumption is more.
CCS chose more LVT cells than in NLDM